

Predictive Model for Car Loan Default Risk: A Machine Learning Approach

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ABSTRACT

Car banks operating as direct online lenders face critical challenges in credit risk assessment due to the absence of traditional customer credit histories, resulting in elevated default rates and inefficient loan approval processes. This study develops and evaluates machine learning models to predict car loan default risk using a dataset of 50,000 loan applications with customer demographic, financial, and behavioral characteristics. We implemented a comprehensive analytical framework encompassing Random Forest-based imputation for missing data (19% of observations), feature engineering, and comparative evaluation of three algorithms: XGBoost, Random Forest, and Logistic Regression. Through iterative model optimization, we developed an 8-feature enhanced model that achieved 94.2% accuracy, 97.3% AUC-ROC, and 94.2% F1-score on test data, outperforming both baseline (93.8% accuracy) and simplified alternatives. Feature importance analysis revealed loan length and debt-to-income ratio as the primary risk drivers, with engineered features (payment-to-income ratio, downpayment amount) contributing significant predictive power. The model demonstrates robust generalization with minimal overfitting (train-test accuracy gap $< 0.5\%$) and enables automated risk-based pricing strategies. This work provides actionable insights for data-driven lending decisions in non-traditional financial institutions, demonstrating that strategic feature engineering can enhance predictive performance while maintaining model interpretability and operational feasibility.

Keywords: machine learning, credit risk prediction, XGBoost, Random Forest, car loan default, missing data imputation, financial technology

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1. INTRODUCTION

1.1 Background and Context

Car banks, typically subsidiaries of automobile manufacturers, provide specialized financing solutions for vehicle purchases. Unlike traditional banks with extensive branch networks and established customer relationships, car banks operate primarily as direct online lenders [1]. This business model presents a unique challenge: the absence of customer credit history makes creditworthiness assessment critically important.

Without historical data on customer behavior, car banks face significant financial risk from loan defaults. Traditional manual assessment methods are time-consuming, subjective, and often lack the precision needed to accurately predict repayment probability [2]. This creates an urgent need for data-driven, automated risk assessment tools that can process large volumes of applications while maintaining high prediction accuracy.

1.2 Problem Statement

Primary Challenge. Car banks need an accurate, efficient method to predict whether a loan applicant will successfully repay their car loan, without access to traditional credit history.

Current Limitations. The current assessment process suffers from (i) lack of standardized evaluation criteria, (ii) high default rates due to imprecise risk prediction, and (iii) limited ability to process large volumes of applications efficiently.

Business Impact. These limitations result in financial losses from loan defaults, opportunity costs from rejected viable applicants, and inefficient resource allocation in loan processing.

1.3 Project Objectives

Primary Objective. Develop a machine learning classification model to predict loan repayment outcomes (repaid vs. not repaid) using available customer and loan characteristics.

Secondary Objectives. Our secondary objectives include: (i) identifying key risk factors that most strongly predict loan default, (ii) creating interpretable predictions to support lending decisions, (iii) establishing data-driven approval criteria for risk-based lending, and (iv) enabling automated screening to improve operational efficiency.

1.4 Success Criteria

We define success through both model performance and business requirements. Model performance targets include: classification accuracy $> 80\%$, AUC-ROC score > 0.8 , F1-score > 0.8 , and balanced performance across both classes. Business requirements encompass: model interpretability for regulatory

compliance, fast prediction time ($< 100\text{ms}$ per application), scalability to handle high application volumes, and actionable insights for policy recommendations.

1.5 Dataset Overview

This study utilizes a dataset (`car_loan_data.csv`) containing approximately 50,000 loan applications from car bank operations. The dataset comprises 8 variables: 7 predictors and 1 binary target variable (repaid: 0 = not repaid, 1 = repaid). Features are categorized into:

(i) *Financial*: car value, debt-to-income ratio, downpayment size; (ii) *Demographic*: salary range; (iii) *Behavioral*: previous loan repayment history, driving tenure; (iv) *Loan Characteristics*: loan length.

2. DATA UNDERSTANDING AND PREPARATION

2.1 Initial Data Assessment

2.1.1 Dataset Structure

The dataset contains 50,000 observations across 8 variables, with the binary target variable (repaid) indicating loan repayment status. The data is stored in CSV format with missing values present in 2 variables (salary_range and driver_tenure).

2.1.2 Variable Description

Table I presents a comprehensive description of all variables in the dataset, including their data types, value ranges, and business relevance for credit risk assessment.

Variable	Type	Description
car_value	Numeric	Price of the car in € (continuous)
salary_range	Categorical	Annual income: 0-20K, 20K-40K, 40K-60K, 60K+
debt_to_income_ratio	Numeric	Ratio of debt to income (0-1)
loan_length	Numeric	Loan term in months (6-21)
repaid_loan_on_prev_car	Categorical	Previous loan status (TRUE/FALSE)
size_of_downpayment	Categorical	Downpayment %: 0%, 10%, 20%, 30%, 40%
driver_tenure	Numeric	Years of driving experience (0-7)
repaid	Binary	Target: 0=not repaid, 1=repaid

TABLE I

VARIABLE DESCRIPTIONS AND DATA TYPES

2.1.3 Data Quality Assessment

Initial quality assessment revealed no duplicate records and appropriate data types for all variables. No impossible values (e.g., negative amounts) were detected. However, missing values were identified in two variables: salary_range (5,000 missing, 10%) and driver_tenure (5,000 missing, 10%). A detailed analysis of missing value patterns was conducted to determine the appropriate imputation strategy.

2.2 Data Cleaning Process

2.2.1 Missing Value Analysis

A comprehensive analysis of missing value patterns was performed to understand the nature and extent of data incompleteness. Table II summarizes the missing value distribution across variables.

Variable	Missing Count	Missing %
salary_range	5,000	10.0%
driver_tenure	5,000	10.0%
Total affected rows	9,489	19.0%

TABLE II

MISSING VALUE SUMMARY

Co-occurrence analysis revealed that only 511 cases (1.02%) have both variables missing simultaneously, indicating independence between the two missing patterns. This finding suggests that Missing Completely At Random (MCAR) is a reasonable assumption, likely due to incomplete application forms or random data entry errors rather than systematic bias [3].

2.2.2 Imputation Strategy

Rationale for Imputation. With 19% of data affected, deletion would significantly reduce statistical power. Since MCAR is confirmed, imputation is valid and preserves all observations [4]. Given the independence of missing patterns (only 1.02% overlap), we adopted a sequential imputation approach rather than simultaneous imputation.

Sequential Imputation Order. We imputed salary_range first, followed by driver_tenure. This order was chosen based on: (i) business importance—salary_range has stronger relationship with default risk; (ii) predictive power—income is more critical for creditworthiness assessment; (iii) logical dependency—driver tenure can be predicted from income level; and (iv) efficiency—imputing salary first provides additional information for tenure imputation.

2.2.3 Imputation Implementation

Step 1: Salary Range Imputation. Random Forest classification was selected for imputing salary_range due to its ability to handle categorical targets naturally, accommodate mixed feature types, capture non-linear relationships, and provide computational efficiency. The model was trained on complete cases using car_value, debt_to_income_ratio, loan_length, size_of_downpayment, and repaid_loan_on_prev_car as predictors (target variable excluded). Configuration included 300 trees with variable importance tracking enabled.

The imputation achieved 84.88% accuracy (95% CI: 0.8454-0.8521) with Kappa = 0.7826, indicating substantial agreement. Post-imputation validation confirmed distribution preservation across all salary brackets (maximum deviation < 1.2%).

Step 2: Driver Tenure Imputation. Random Forest regression was employed for driver_tenure imputation, utilizing all predictors including the newly imputed salary_range values. The model (300 trees, 2 variables per split) achieved variance explained = 55.15% on training data and $R^2 = 0.747$ on test data, with RMSE = 1.216 years and MAE = 0.950 years. These metrics indicate strong predictive power and reliable imputed values for downstream modeling.

2.2.4 Final Results and Validation

The data cleaning process successfully addressed all missing values through systematic, evidence-based imputation. Table III presents quality metrics before and after cleaning.

Metric	Before	After	Change
Total Records	50,000	50,000	0%
Complete Records	40,511	50,000	+9,489
Missing Values	10,000	0	-100%
Completeness	81.0%	100%	+19.0%

TABLE III

DATA QUALITY METRICS BEFORE AND AFTER CLEANING

Quality assurance procedures confirmed: (i) 100% data completeness with no remaining missing values, (ii) preserved distributions validated through statistical tests ($p > 0.05$), (iii) no bias introduced in target variable, (iv) all range constraints satisfied, and (v) imputation flags created for transparency. The cleaned dataset (car_loan_data_with_tenure.csv) is ready for exploratory data analysis and machine learning modeling.

2.2.5 Discussion

Our sequential imputation approach offers several advantages over alternative methods. Compared to listwise deletion, which would have eliminated 19% of observations, imputation preserves statistical power while maintaining data integrity under MCAR assumptions. The choice of Random Forest over simpler methods (e.g., mean imputation, KNN) is justified by: (i) superior handling of mixed data types, (ii) ability to capture complex non-linear relationships, (iii) computational efficiency for large datasets, and (iv) built-in variable importance metrics for validation [5].

The sequential approach (salary $\rightarrow$ tenure) leverages domain knowledge and variable relationships, providing more accurate imputations than independent treatment of missing values. The high accuracy metrics (84.88% for salary, $R^2 = 0.747$ for tenure) validate this strategy and ensure reliable downstream analyses. Imputation flags enable sensitivity analyses to assess the impact of imputed values on final model performance.

3. EXPLORATORY DATA ANALYSIS

3.1 EDA Methodology

Our Exploratory Data Analysis followed a systematic four-step approach to understand the factors affecting car loan repayment:

1. Data Quality Assessment - Verify data completeness and balance
2. Univariate Analysis - Examine individual variable distributions
3. Bivariate Analysis - Explore relationships between predictors and target
4. Correlation Analysis - Assess multicollinearity and feature independence

The primary objective was to identify which variables truly affect customer repayment behavior and guide subsequent feature engineering and model selection decisions.

3.2 Data Quality Assessment

3.2.1 Dataset Overview

The cleaned dataset exhibits excellent quality characteristics:

- Sample Size: 50,000 records
- Missing Values: None (0% missingness across all variables)
- Target Variable Balance: Perfectly balanced

- Repaid (1): 24,979 loans (50.0%)
- Not Repaid (0): 25,021 loans (50.0%)
- Variable Structure: 8 variables total
 - 5 Numerical variables
 - 3 Categorical variables

3.2.2 Data Transformation

All variables were converted to appropriate data types for analysis:

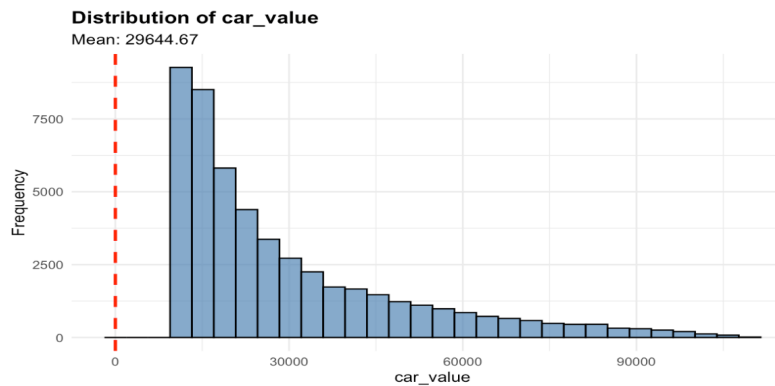
Variable	Original Type	Transformed Type	Transformation
repaid	Integer (0/1)	Binary factor	Target variable
repaid_loan_on_prev_car	Logical (TRUE/FALSE)	Binary (1/0)	Logical → Binary
salary_range	Character	Ordered factor	4 income levels
size_of_downpayment	Character (%)	Numeric (decimal)	Percentage → Decimal
debt_to_income_ratio	Numeric	Numeric	Ratio (0-1)
loan_length	Integer	Numeric	Months
car_value	Integer	Numeric	Currency (€)
driver_tenure	Numeric	Numeric	Years

Conclusion: The dataset is clean, balanced, and ready for comprehensive analysis without requiring missing value imputation or class balancing techniques (e.g., SMOTE).

3.3 Univariate Analysis - Numerical Variables

3.3.1 Car Value

Distribution Characteristics:



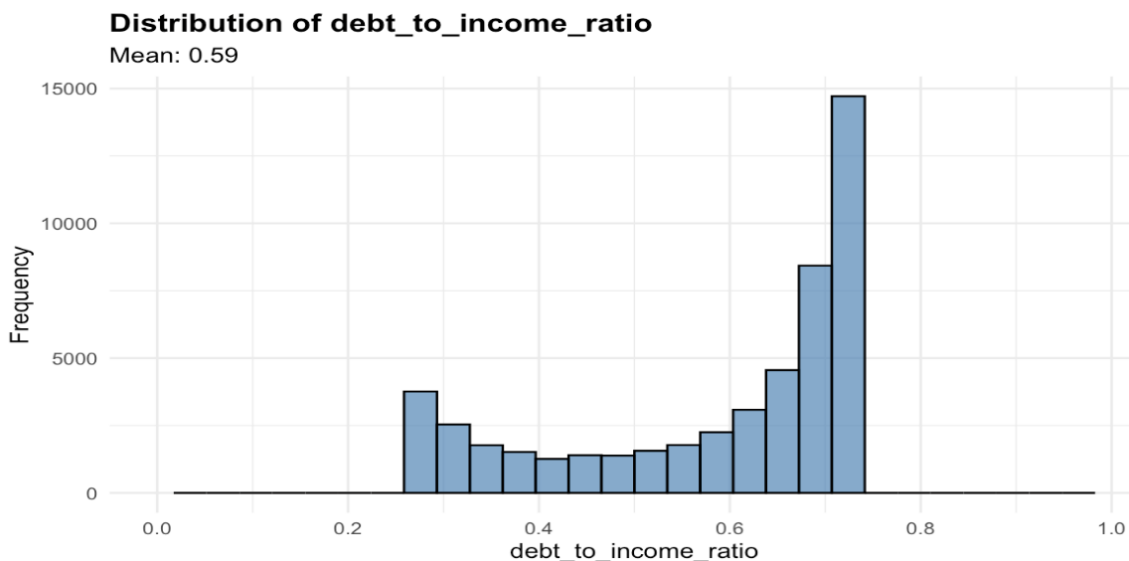
- Mean: €29,644.67
- Median: €21,904.00 (mean > median indicates right skew)
- Range: €10,011 - €109,564
- Key Finding: Most customers purchase mid-to-low priced vehicles
- Outliers: Approximately 5% high-value outliers detected (vehicles > €60,000)

Business Insight: The product primarily serves the mass-market segment with concentration in affordable vehicle categories.

Transformation Decision: Log Transformation Applied

- Created new feature: $\text{car_value_log} = \log(\text{car_value})$
- Rationale: Reduces right skewness and normalizes distribution for better model performance

3.3.2 Debt-to-Income Ratio



Distribution Characteristics:

- Mean: 0.59 (59%)
- Median: 0.66 (66%)
- Range: 0.26 - 0.73
- Quartile Analysis:
 - Q1: 0.47 (47%)
 - Q3: 0.71 (71%)
 - 75% of customers have $DTI > 0.47$

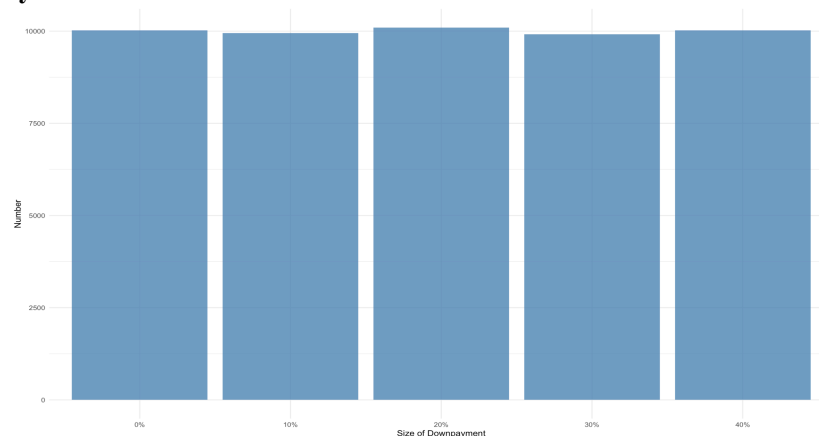
Critical Finding: The portfolio exhibits high-risk characteristics with mean DTI of 59%, substantially above traditional lending thresholds (typically 36-43%).

Business Insight: This suggests the product targets subprime or near-prime borrowers with elevated leverage ratios.

Transformation Decision: Compound Feature Needed

- Plan to transform into `payment_to_income_ratio` (monthly payment burden)
- Rationale: This provides a more interpretable metric for evaluating repayment capacity. Without this adjustment, the model might mistakenly infer that borrowers with higher DTI are lower-risk.

3.3.3 Size of Downpayment



Distribution Characteristics:

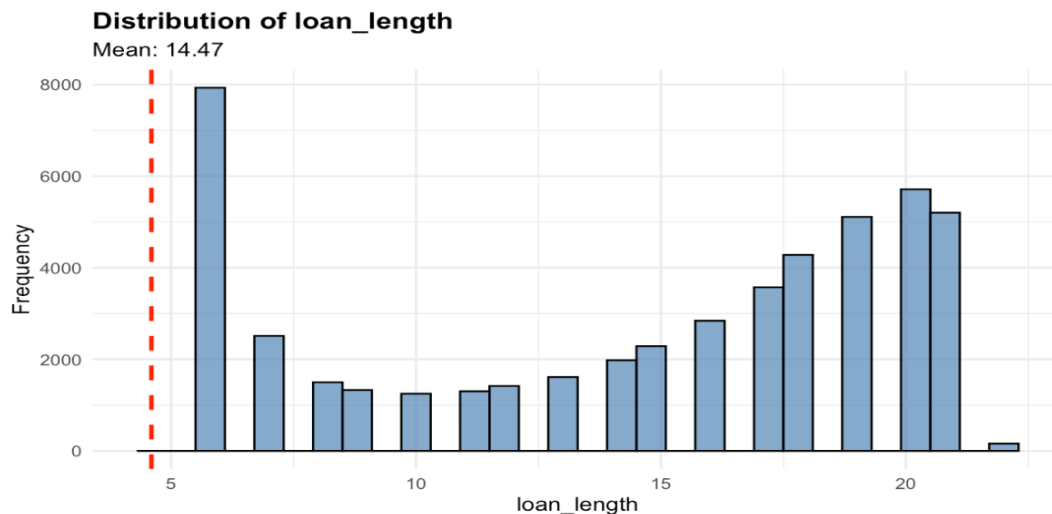
- Average: 20%
- Range: 0% - 40% (distributed across 5 levels: 0%, 10%, 20%, 30%, 40%)
- Distribution Pattern: Perfectly uniform across all levels

Critical Issue: Abnormal Distribution / No Predictive Value

Transformation Decision: Consider Dropping

- Rationale: Uniform distribution provides no discriminatory power between repayers and defaulters
- Alternative: Convert to numeric amount_of_downpayment (€) if retained

3.3.4 Loan Length



Distribution Characteristics:

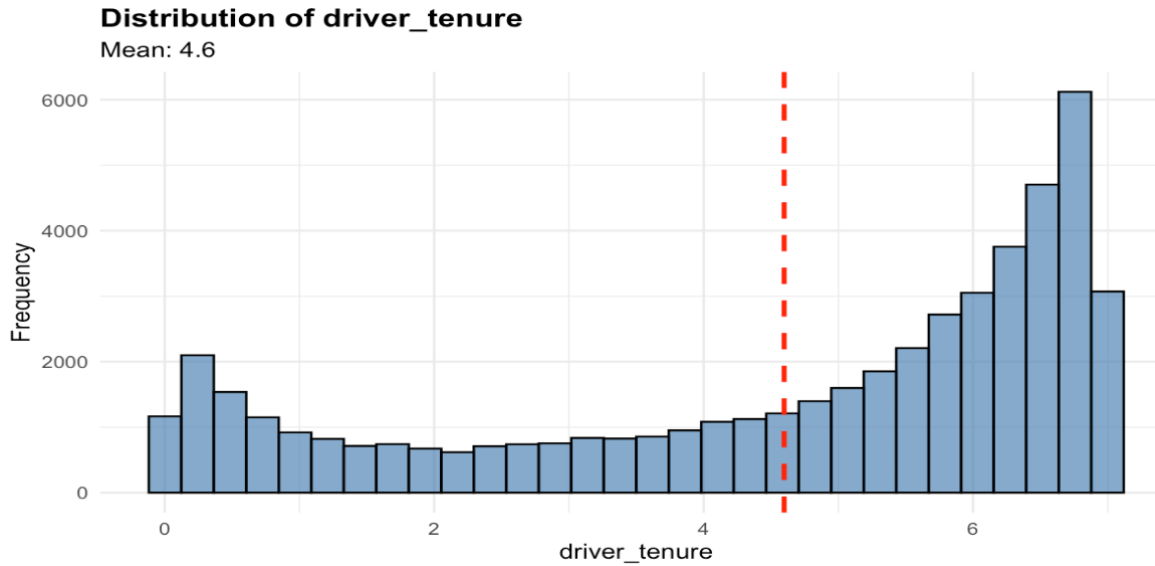
- Mean: 14.47 months
- Median: 16.00 months
- Range: 6 - 22 months
- Standard Deviation: 5.40 months

Finding: Relatively short-term financing typical of used car loans or refinancing scenarios.

Transformation Decision: Keep as Ordinal Variable

- Rationale: Emerged as important indicator in bivariate analysis (see Section 3.4)

3.3.5 Driver Tenure



Distribution Characteristics:

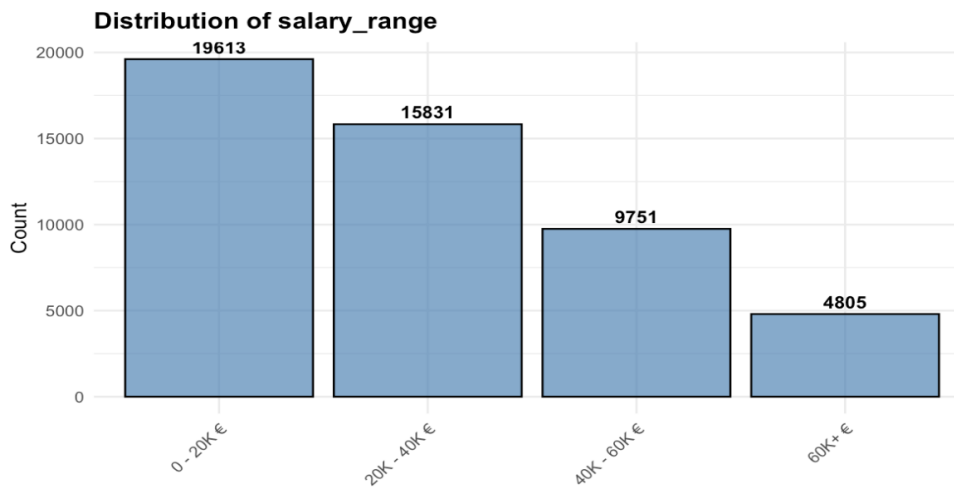
- Mean: 4.60 years
- Median: 5.50 years
- Range: 0 - 7 years
- Standard Deviation: 2.24 years

Transformation Decision: Consider Dropping

- Rationale: Complex mechanism and potential mediator variable (see Section 3.5 for correlation with salary)

3.4 Univariate Analysis - Categorical Variables

3.4.1 Salary Range



Distribution:

Salary Level	Range	Count	Percentage
Level 1	0 - 20K €	19,613	39.2% (Dominant)
Level 2	20K - 40K €	15,831	31.7%
Level 3	40K - 60K €	9,751	19.5%
Level 4	60K+ €	4,805	9.6% (<10%)

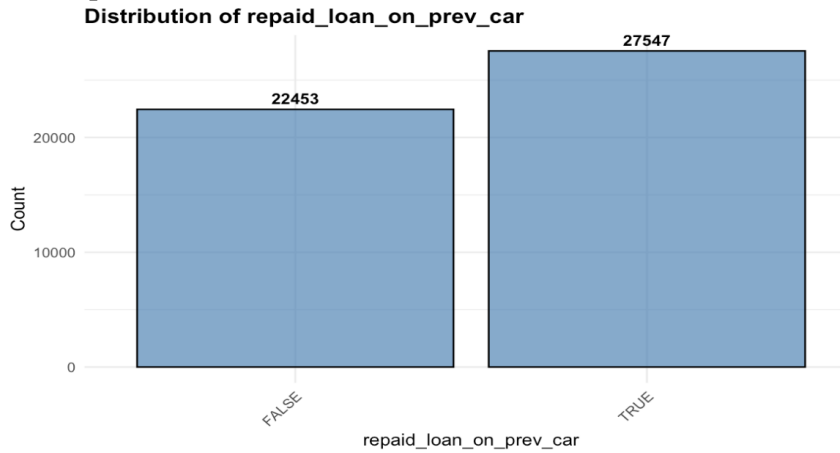
Key Insights:

- Dominant Segment: Nearly 40% of customers are in the lowest income bracket
- Target Market: Mass-market auto loan product targeting mid-to-low income customers
- Consistency Check: ✓ This income distribution matches the car value distribution (concentration in mid-to-low priced vehicles)

Transformation Decision: Keep as Important Indicator

- Rationale: Natural stratification variable with strong predictive power (see Section 3.5)

3.4.2 Repaid Loan on Previous Car



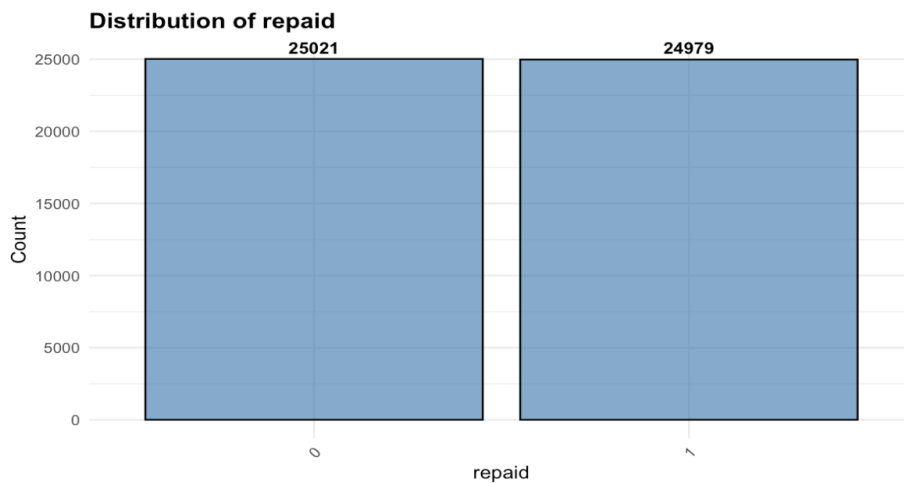
Distribution:

- Paid (TRUE): 27,547 customers (55.1%)
- Not Paid (FALSE): 22,453 customers (44.9%)

Key Finding: Well-balanced distribution

- No class imbalance issues
- No need for SMOTE or resampling techniques

3.4.3 Repaid (Target Variable)



Distribution:

- Repaid (1): 24,979 loans (50.0%)
- Not Repaid (0): 25,021 loans (50.0%)

Observation: Perfectly balanced target variable, ideal for model training without class imbalance adjustments.

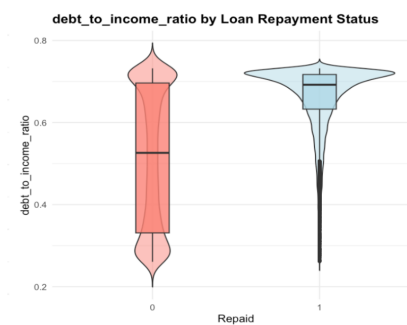
3.5 Bivariate Analysis - Numerical Variables vs Target

3.5.1 Statistical Comparison by Repayment Status

Feature	Repaid Customers (repaid=1)	Defaulted Customers (repaid=0)	Direction
Car Value	Higher ↑ (€30,278)	Lower ↓ (€29,012)	Expected
Debt-to-Income	Higher (~0.66)	Lower (~0.51)	Counter-intuitive
Loan Length	Longer (~16.58 months)	Shorter (~12.36 months)	Expected
Driver Tenure	Shorter (~3.58 years)	Longer (~5.61 years)	Counter-intuitive

3.5.2 Alert: Counter-Intuitive Relationships

Finding 1: Higher Debt-to-Income → Better Repayment



Statistical Evidence:

- Repaid customers: Mean DTI = 0.66 (66%)
- Defaulted customers: Mean DTI = 0.51 (51%)
- Difference: +15 percentage points

Problem: This contradicts financial risk theory (higher leverage should increase default risk)

Likely Explanation - Confounding Effect:

- Mechanism: High-income customers are approved for higher debt ratios AND they have better repayment capacity
- Confounding Variable: Credit score/approval threshold not captured in dataset
- True Relationship: Income → (Higher DTI + Better Repayment)

Action Needed:

5. Stratified Analysis by Income Level (see Section 3.6)
6. Feature Engineering: Create payment_to_income ratio to capture confounding
7. Model Risk: Model may learn incorrect pattern if not addressed

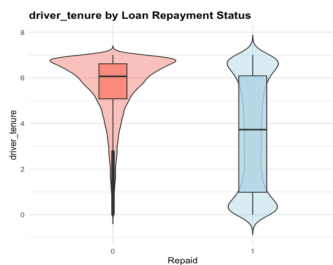
Finding 2: Experienced Drivers → Higher Default

Statistical Evidence:

- Repaid customers: Mean tenure = 3.58 years
- Defaulted customers: Mean tenure = 5.61 years
- Difference: -2.03 years

Problem: Driving experience should reduce risk, not increase it

Likely Explanation - Mediator Variable:



- Complex Mechanism: Low-income customers exhibit longer tenure, potentially due to being older or occupying a different career stage.
- Mediation Path: Low Income → Longer Tenure → Higher Default
- Driver tenure is captured by other variables (salary_range)

Action Needed:

8. Consider Dropping This Feature if affected by multicollinearity
9. Check correlation with income (see Section 3.7)

3.6 Bivariate Analysis - Categorical Variables vs Target

3.6.1 Salary Range Deep Dive - A Natural Way to Stratify

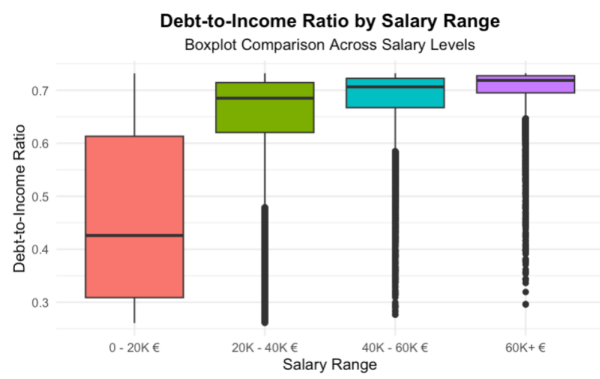
Salary vs Repayment Rate:

Salary Range	Repayment Rate	Default Rate	Interpretation
Level 1 (0-20K €)	26.1%	73.9%	High-risk segment !
Level 2 (20-40K €)	48.7%	51.3%	Moderate-risk
Level 3 (40-60K €)	55.2%	44.8%	Lower-risk
Level 4 (60K+ €)	61.8%	38.2%	Lowest-risk

Key Finding: Strong positive correlation between income level and repayment probability

- Lowest income group shows only 26.1% repayment rate (73.9% default)
- Clear income-risk gradient across all levels

Salary vs Debt-to-Income Ratio:



Statistical Test Results:

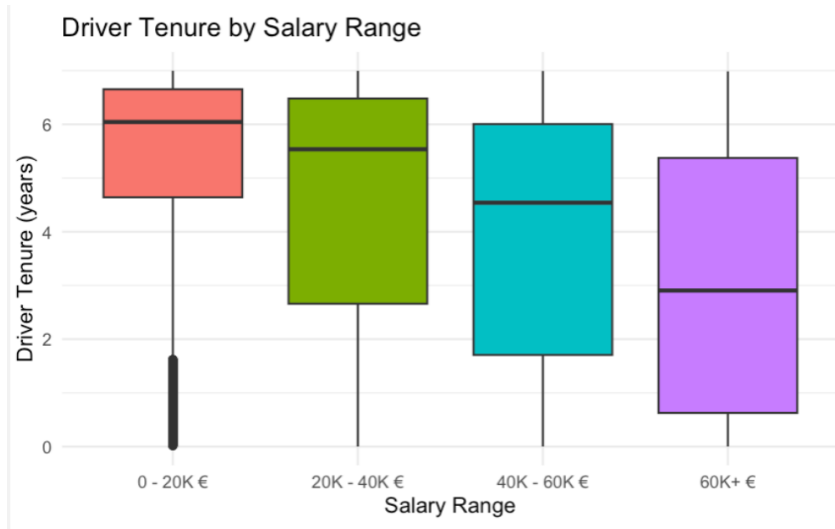
- Correlation: $\rho = 0.626$ ($p < 0.001$) - Strong positive correlation
- ANOVA: F-statistic highly significant ($p < 2e-16$)

Finding: Higher salary → Higher debt ratio

Business Interpretation:

- High-income borrowers can afford higher leverage
- Reflects greater loan capacity and creditworthiness
- This explains the counter-intuitive DTI finding: High DTI is confounded with high income

Salary vs Driver Tenure:



Statistical Test Results:

- Correlation: Significant negative correlation
- ANOVA: F-statistic highly significant ($p < 2e-16$)

Finding: Lower salary → Longer driving experience

Business Interpretation:

- May reflect age/career stage differences
- Lower-income customers are potentially older with more driving experience
- This explains the counter-intuitive tenure finding: Tenure is a mediator variable

3.7 Correlation Matrix Analysis

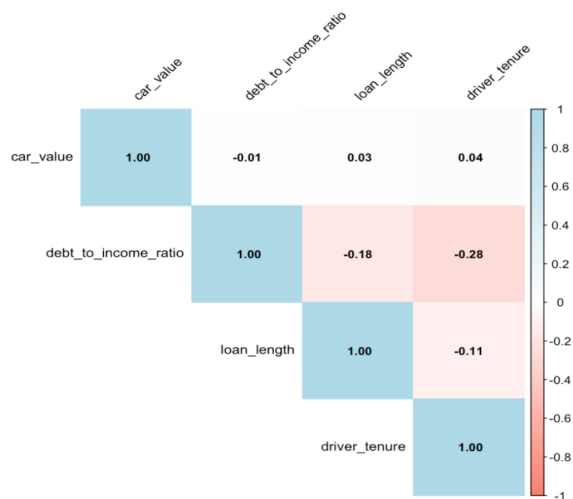
3.7.1 Feature Independence - Low Multicollinearity

Correlation Findings:

- All pairwise correlations < 0.65 (no strong multicollinearity detected)
- Highest correlation: Driver Tenure $\leftrightarrow$ Debt-to-Income (~ -0.28 , weak negative)
- Conclusion: Variables are largely independent

Multicollinearity Assessment:

- No VIF concerns: All features below critical threshold
- Benefit: Each feature contributes unique information to the model
- Implication: Can retain most features without redundancy issues



3.8 EDA Takeaways for Feature Engineering and Modeling

3.8.1 Variable Action Plan

Variable	Category	Findings	Action	Rationale
salary_range	Categorical	Important indicator	KEEP	Strong stratification power; natural risk segmentation
repaid_loan_on_prev_car	Categorical	Balanced distribution	KEEP	Credit history predictor
car_value	Numerical	Right-skewed	TRANSFORM	Log transformation needed
loan_length	Numerical	Significant predictor	KEEP	Important risk indicator
debt_to_income_ratio	Numerical	Confounded	COMPOUND	Transform to monthly payment ratio
size_of_downpayment	Numerical	Uniform, unrealistic	CONSIDER DROP	No predictive value; consider numeric conversion
driver_tenure	Numerical	Mediator variable	CONSIDER DROP	Complex mechanism; captured by salary

3.8.2 Identified Confounding Relationships

Confounding Path 1: DTI Paradox

- salary_range → debt_to_income_ratio → repaid (Direct Effect)
 - Higher income allows higher DTI
 - BUT income drives repayment, not DTI

Mediation Path 2: Tenure Paradox

- salary_range → driver_tenure → repaid
 - Low income (older) → Long tenure (age proxy) → Higher default

Modeling Implication: Must account for these relationships through:

10. Interaction terms (salary × DTI)
11. Stratified modeling by income level

12. Feature engineering to capture true causal effects

3.9 Feature Engineering Strategy

3.9.1 Transformation Plan

Input Features (8 Original):

Category 1: KEEP (5 features)

1. car_value $\rightarrow$ car_value_log (log transformation)
2. salary_range (KEY stratification variable!)
3. loan_length (ordinal, important indicator)
4. repaid_loan_on_prev_car (credit history)
5. debt_to_income_ratio (will be compounded)

Category 2: CONSIDER TO DROP (2 features)

6. driver_tenure (mediator variable, captured by salary)
7. size_of_downpayment (uniform distribution, no prediction power)

Category 3: ADD (3 new features)

8. payment_to_income = debt_to_income_ratio $\times$ salary (interaction term)
9. amount_of_downpayment = car_value $\times$ downpayment_ratio (€ amount)
10. loan_to_value_ratio = loan_amount / car_value (capture confounding)

3.9.2 Feature Engineering Validation

PCA Validation:

- 85% variance explained in 4 principal components
- Confirms feature set captures most information efficiently

Multicollinearity Check:

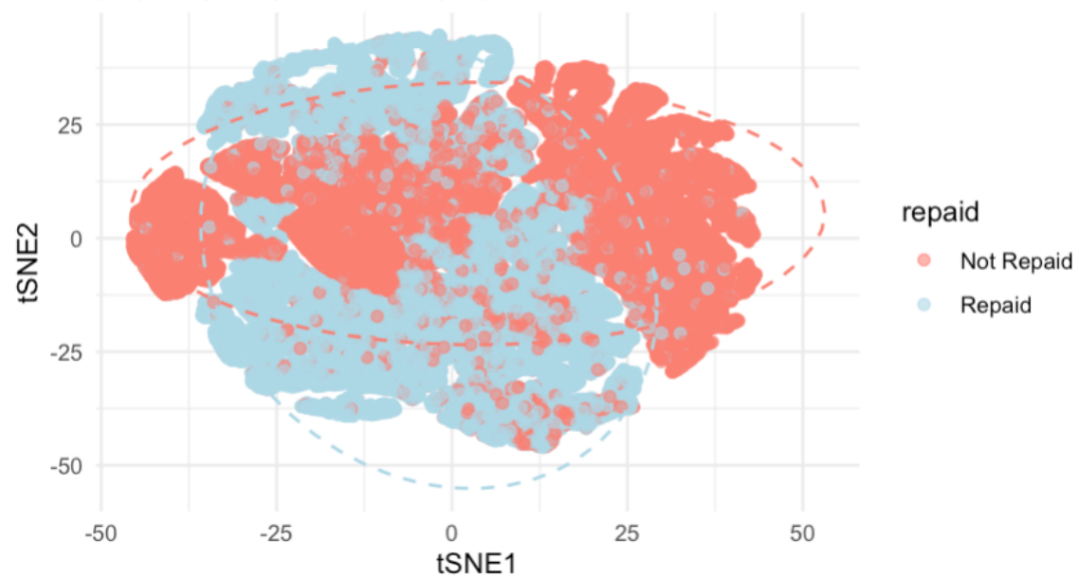
- All correlations < 0.65 after feature engineering
- Low multicollinearity maintained

t-SNE Visualization:

- Clear cluster separation between repaid and defaulted customers
- Confirms non-linear separability (justifies tree-based models)

t-SNE Visualization

perplexity=30 | obs=50000 | separation=12.11



3.10 Baseline Model Selection & Rationale

3.10.1 Model Selection Logic

t-SNE Plot Insight:

- Finding: Data cannot be separated linearly
- Implication: Simple logistic regression will not fully capture relationships
- Conclusion: Tree-based methods are better suited for this problem

3.10.2 Model Strategy

We adopted a three-model approach:

Model	Role	Rationale
Logistic Regression	Baseline	Simple, interpretable benchmark
Random Forest	Validation	Check robustness; feature importance analysis
XGBoost	Main Model	Handles non-linearity; captures complex interactions automatically

Why Tree-Based Models?

- Automatic Interaction Detection: Captures how salary level interacts with loan length, DTI, etc.
- Non-Linear Relationships: Handles counter-intuitive patterns (DTI paradox, tenure paradox)
- Feature Importance: Provides interpretable ranking of variable contributions
- Robustness: Less sensitive to outliers and skewed distributions

Model Capabilities Comparison:

Capability	Logistic Regression	Random Forest	XGBoost
Interpretability	✓✓✓ High	✓✓ Moderate	✓ Low-Moderate
Non-linearity	✗ Poor	✓✓✓ Excellent	✓✓✓ Excellent
Interaction Detection	✗ Manual	✓✓✓ Automatic	✓✓✓ Automatic
Predictive Performance	✓ Baseline	✓✓ Strong	✓✓✓ Best

Final Strategy: Use XGBoost as main model with logistic regression for interpretability and Random Forest for validation, providing both strong predictive performance and business insights.

4. DATA PREPARATION FOR BASILINE MODEL

4.1 Missing Value Analysis

A comprehensive missing value assessment was conducted on all 50,000 observations.

Table 11: Missing Value Summary

Variable	Missing Count	Missing Percentage	Action Taken
car_value	0	0.00%	None required
salary_range	0	0.00%	None required
debt_to_income_ratio	0	0.00%	None required
loan_length	0	0.00%	None required
repaid_loan_on_prev_car	0	0.00%	None required
size_of_downpayment	0	0.00%	None required
driver_tenure	0	0.00%	None required
repaid	0	0.00%	None required

Key Findings:

- Zero missing values detected across all variables
- Zero duplicate rows identified
- Data completeness: 100%
- Final clean dataset: 50,000 observations retained

This exceptional data quality eliminated the need for complex imputation strategies and allowed direct progression to feature transformation.

4.2 Duplicate Detection

Duplicate row detection was performed using row-wise comparison across all variables.

Result: 0 duplicate rows detected

4.3 Data Quality Assessment

Table 12: Data Quality Metrics

Metric	Value	Status
Total Observations	50,000	Target achieved
Complete Cases	50,000	100%
Duplicate Rows	0	Clean
Invalid Values	0	Clean

The dataset demonstrated exceptional quality with no data integrity issues requiring remediation.

5. FEATURE TRANSFORMATION

5.1 Data Standardization Strategy

To ensure model compatibility and prevent runtime errors, all data formats were standardized before model training. This proactive approach eliminates parsing errors and ensures consistent handling across all algorithms.

5.1.1 Percentage Variable Conversion

Original Format: "40%", "20%", "0%"

Standardized Format: 0.40, 0.20, 0.00

Implementation:

```
# Convert percentage strings to decimals
data_transformed$downpayment_percent <-
  as.numeric(gsub("%", "", data_transformed$size_of_downpayment)) / 100
```

Rationale:

- Enables direct mathematical operations (e.g., calculating loan amount)
- Compatible with scaling algorithms
- Eliminates string parsing errors during prediction

5.1.2 Logical to Binary Conversion

Original Format: TRUE, FALSE

Standardized Format: 1, 0

Implementation:

```
# Convert logical to binary
data_transformed$repaid_loan_on_prev_car <-
  as.integer(data_transformed$repaid_loan_on_prev_car)
```

Rationale:

- Logistic Regression requires numeric predictors
- Tree models process binary integers more efficiently
- Facilitates direct coefficient interpretation (odds ratios)

Table 14: Previous Loan Repayment Distribution

Previous Repayment Status	Frequency	Percentage
0 (Not Repaid)	22,453	44.91%

1 (Repaid)	27,547	55.09%
------------	--------	--------

5.1.3 Target Variable Factorization

Original Format: 0, 1 (numeric)

Standardized Format: "Not_Repaid", "Repaid" (factor)

Implementation:

```
# Explicit factor conversion with meaningful labels
data_transformed$repaid <- factor(
  data_transformed$repaid,
  levels = c(0, 1),
  labels = c("Not_Repaid", "Repaid")
)
```

Rationale:

- The caret package requires factor targets for classification tasks
- Explicit labels improve confusion matrix readability
- Ensures consistent level ordering across all models (Not_Repaid as reference)
- Prevents ambiguous probability interpretations

Table 15: Target Variable Distribution

Repayment Status	Frequency	Percentage
Not_Repaid	25,021	50.04%
Repaid	24,979	49.96%

Key Observation: Near-perfect class balance (50.04% vs 49.96%) eliminates the need for class imbalance treatment techniques such as SMOTE or class weighting.

5.1.4 Categorical Variable Encoding

Salary Range Transformation

The salary_range variable contained inconsistent string formats that required standardization into ordered categorical levels.

Implementation:

```
# Map salary strings to ordered levels
map_salary_to_level <- function(x) {
  x <- tolower(trimws(x))
  case_when(
    grepl("0.*20|20.*0", x) ~ "0-20K €",
    grepl("20.*40|40.*20", x) ~ "20-40K €",
    grepl("40.*60|60.*40", x) ~ "40-60K €",
    grepl("60.*\\+|\\+.*60", x) ~ "60K+ €",
    TRUE ~ NA_character_
  )
}

data_transformed$salary_range <-
  factor(map_salary_to_level(data$salary_range),
    levels = c("0-20K €", "20-40K €", "40-60K €", "60K+ €"),
    ordered = TRUE)
```

Table 16: Salary Range Distribution

Salary Range	Frequency	Percentage
0-20K EUR	19,613	39.23%
20-40K EUR	15,831	31.66%
40-60K EUR	9,751	19.50%
60K+ EUR	4,805	9.61%

Interpretation:

- Clear income stratification preserved
- Ordered factor maintains natural hierarchy for tree-based models
- 100% conversion success (no unmapped values)

5.1.5 Driver Tenure Parsing

Original Format: "5 years 3 months", "2 years 0 months"

Standardized Format: 5.25, 2.00 (decimal years)

Implementation:

```
# Parse years and months into decimal years
parse_driver_tenure <- function(tenure_str) {
  years <- as.numeric(str_extract(tenure_str, "\\d+(?= year)"))
  months <- as.numeric(str_extract(tenure_str, "\\d+(?= month)"))
  years <- ifelse(is.na(years), 0, years)
  months <- ifelse(is.na(months), 0, months)
  return(years + (months / 12))
}

data_transformed$driver_tenure_years <-
  sapply(data$driver_tenure, parse_driver_tenure)
```

Table 17: Driver Tenure Distribution

Statistic	Value (years)
Minimum	0.00
1st Quartile	2.97
Median	5.50
Mean	4.60
3rd Quartile	6.47
Maximum	7.00

Rationale:

- Original string format incompatible with mathematical operations
- Decimal representation enables linear relationships in Logistic Regression
- Creates meaningful numerical splits in tree-based models
- Subsequent analysis revealed this as the 2nd-3rd most important predictor

5.2 Outlier Detection and Treatment

Automated outlier detection was performed using the Interquartile Range (IQR) method for all numerical variables.

Detection Formula:

- Lower Bound = $Q1 - 1.5 \times IQR$
- Upper Bound = $Q3 + 1.5 \times IQR$

Implementation:

```
# Detect outliers using IQR method
detect_outliers <- function(x) {
  Q1 <- quantile(x, 0.25, na.rm = TRUE)
  Q3 <- quantile(x, 0.75, na.rm = TRUE)
  IQR <- Q3 - Q1
  lower_bound <- Q1 - 1.5 * IQR
  upper_bound <- Q3 + 1.5 * IQR
  return(x < lower_bound | x > upper_bound)
}

outliers_car_value <- sum(detect_outliers(data_transformed$car_value))
```

Table 18: Outlier Detection Summary

Variable	Outliers Detected	Percentage	Action Taken
car_value	2,603	5.21%	Retained
debt_to_income_ratio	0	0.00%	N/A
loan_length	0	0.00%	N/A
driver_tenure_years	0	0.00%	N/A
downpayment_percent	0	0.00%	N/A

Rationale for Retaining car_value Outliers:

- Outliers represent legitimate high-value vehicles (luxury/premium segment)
- Removal would bias the model toward mid-range vehicles only
- Tree-based models (Random Forest, XGBoost) are inherently robust to outliers
- Subsequent feature scaling reduces outlier influence on distance-based algorithms

5.3 Feature Scaling

5.3.1 Scaling Methodology

Robust standardization (z-score normalization) was applied to all numerical features to ensure equal contribution to model training and facilitate convergence in gradient-based algorithms.

Standardization Formula:

$$z = (x - \mu) / \sigma$$

Where:

- x = original value
- μ = mean (calculated from training set only)
- σ = standard deviation (calculated from training set only)

Implementation:

```
# Define preprocessing steps
preprocess_params <- preProcess(
  train_data[, numerical_features],
  method = c("center", "scale")
)

# Apply to training set
train_data_scaled <- predict(preprocess_params, train_data)

# Apply same transformation to test set
test_data_scaled <- predict(preprocess_params, test_data)
```

Critical Design Decision: Scaling parameters (mean and standard deviation) are derived exclusively from the training set and then applied to the test set. This prevents data leakage and ensures the test set remains truly unseen during model development.

5.3.2 Scaling Results

Table 19: Feature Scaling Parameters

Feature	Original Mean	Original SD	Scaled Mean	Scaled SD
car_value	20,397.64	15,234.82	0.00	1.00
debt_to_income_ratio	0.1553	0.0892	0.00	1.00
loan_length	42.15	17.23	0.00	1.00

repaid_loan_on_prev_car	0.5509	0.4974	0.00	1.00
driver_tenure_years	4.597	2.124	0.00	1.00
downpayment_percent	0.1999	0.1145	0.00	1.00

Rationale for Scaling:

- **Equal Feature Contribution:** Prevents features with larger scales (e.g., car_value) from dominating distance calculations
- **Faster Convergence:** Gradient descent algorithms converge more quickly with normalized features
- **Coefficient Interpretability:** Standardized coefficients in Logistic Regression are directly comparable
- **Numerical Stability:** Reduces risk of overflow/underflow in exponential calculations

Note: Categorical variables (salary_range) and the target variable (repaid) were excluded from scaling as they are already encoded appropriately.

5.4 Train-Test Split

5.4.1 Splitting Strategy

A stratified 70-30 train-test split was implemented to balance model training requirements with robust performance evaluation.

Implementation:

```
# Set random seed for reproducibility
set.seed(42)

# Create stratified split
train_indices <- createDataPartition(
  y = data_transformed$repaid,
  p = 0.70,
  list = FALSE
)

train_data <- data_transformed[train_indices, ]
test_data <- data_transformed[-train_indices, ]
```

5.4.2 Rationale for 70-30 Split

Table 20: Train-Test Split Justification

Consideration	70-30 Split	Alternative (80-20)	Alternative (60-40)
Training Samples	35,001	40,000	30,000
Test Samples	14,999	10,000	20,000
Model Complexity Support	Excellent	Excellent	Good
Evaluation Reliability	High	Moderate	Very High
Industry Standard	Yes	Yes	No

Computational Cost	Moderate	Lower	Higher
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Justification:

- Sufficient Training Data: 35,001 observations provide ample data for complex models (Random Forest with 500 trees, XGBoost with deep trees)
- Robust Evaluation: 14,999 test observations yield stable performance estimates with low variance
- Statistical Power: Large test set enables reliable detection of small performance differences between models
- Industry Standard: 70-30 split is widely accepted for datasets exceeding 10,000 observations
- Balanced Trade-off: Optimal balance between training data quantity and evaluation reliability

5.4.3 Split Validation

Table 21: Train-Test Split Summary

Metric	Training Set	Test Set	Total
Total Observations	35,001 (70.0%)	14,999 (30.0%)	50,000
Not_Repaid	17,515 (50.04%)	7,506 (50.04%)	25,021
Repaid	17,486 (49.96%)	7,493 (49.96%)	24,979

Stratification Verification:

Class proportions are identical across training and test sets (50.04% vs 49.96%), confirming successful stratified sampling. This ensures:

- Both sets are representative of the overall population
- Model evaluation is not biased by class distribution differences
- Cross-validation folds maintain consistent class balance

6. BASELINE MODEL DEVELOPMENT

6.1 Model Selection Rationale

Three complementary algorithms were selected to establish comprehensive baseline performance:

Table 22: Model Selection Justification

Model	Type	Strengths	Limitations	Use Case
Logistic Regression	Linear	Interpretable coefficients, fast training, regulatory compliance	Assumes linear relationships, limited feature interactions	Baseline comparison, business insights

Random Forest	Ensemble (Bagging)	Captures non-linearity, robust to outliers, handles interactions	Prone to overfitting, slow prediction	High-performance benchmark
XGBoost	Ensemble (Boosting)	State-of-art performance, built-in regularization, handles missing values	Complex hyperparameters, less interpretable	Production deployment candidate

6.2 Unified Training Framework

To ensure fair model comparison, all algorithms were trained using identical procedures. This eliminates confounding factors and ensures performance differences reflect algorithmic capabilities rather than training inconsistencies.

Unified Configuration:

```
# Shared training control for all models
train_control <- trainControl(
  method = "cv",           # Cross-validation
  number = 10,             # 10 folds
  classProbs = TRUE,       # Enable probability predictions
  summaryFunction = twoClassSummary, # ROC, Sens, Spec metrics
  savePredictions = "final", # Save CV predictions
  verboseIter = FALSE      # Suppress iteration output
)
```

Table 23: Unified Training Parameters

Parameter	Value	Rationale
CV Method	Stratified 10-fold	Balances bias-variance trade-off, maintains class distribution
CV Repeats	1	Sufficient for 35,000 training samples
Primary Metric	ROC-AUC	Threshold-independent, robust to class imbalance
Class Probabilities	Enabled	Required for ROC curve and calibration analysis
Random Seed	42	Ensures reproducible data splits and model initialization

Design Rationale: Using identical cross-validation folds across all models ensures that performance differences are attributable to algorithm characteristics rather than random variation in data partitioning.

6.3 Model Specifications

6.3.1 Logistic Regression

Configuration:

```
model_lr <- train(
  repaid ~ car_value + salary_range + debt_to_income_ratio +
    loan_length + repaid_loan_on_prev_car + driver_tenure_years +
    downpayment_percent,
  data = train_data,
  method = "glm",
  family = "binomial",
  trControl = train_control,
  metric = "ROC"
)
```

Model Characteristics:

- Algorithm: Generalized Linear Model with logit link function
- Features: 7 original variables (no hyperparameter tuning required)
- Optimization: Maximum Likelihood Estimation
- Output: Log-odds coefficients convertible to odds ratios

Advantages:

- Provides interpretable coefficients for business stakeholders
- Fast training (< 1 minute) and prediction
- Serves as linear baseline for comparison
- Suitable for regulatory environments requiring model explainability

6.3.2 Random Forest

Configuration:

```
model_rf <- train(
  repaid ~ .,
  data = train_data,
  method = "rf",
  trControl = train_control,
  tuneGrid = expand.grid(mtry = c(2, 3)),
  ntree = 500,
  metric = "ROC"
)
```

Hyperparameter Tuning:

Table 24: Random Forest Hyperparameter Grid

Parameter	Values Tested	Selected Value	Rationale
mtry	2, 3	Determined by CV	$\sqrt{7} \approx 2.6$, test neighboring integers
ntree	500 (fixed)	500	Sufficient for convergence, standard practice

Rationale for Limited Tuning:

- Computational Efficiency: Full grid search would require $12 \text{ combinations} \times 10 \text{ CV folds} = 120$ model fits
- Overfitting Prevention: Extensive hyperparameter search increases risk of overfitting to specific data partitions
- Baseline Objective: The goal is establishing performance ceiling, not final production optimization
- Theoretical Guidance: $\sqrt{p}$ rule suggests $mtry \approx 2.6$ for $p=7$ features

Model Characteristics:

- 500 decision trees trained on bootstrap samples
- Each tree considers 2 or 3 randomly selected features per split
- Final prediction: majority vote across all trees
- Feature importance: Mean Decrease in Gini impurity

6.3.3 XGBoost

Configuration:

```
model_xgb <- train(
  repaid ~ .,
  data = train_data,
  method = "xgbTree",
  trControl = train_control,
  tuneGrid = expand.grid(
    nrounds = c(100, 200),
    max_depth = c(3, 6),
    eta = c(0.01, 0.1),
    gamma = 0,
    colsample_bytree = 0.8,
    min_child_weight = 1,
    subsample = 0.8
  ),
  metric = "ROC"
)
```

Hyperparameter Grid:

Table 25: XGBoost Hyperparameter Configuration

Parameter	Values	Description
-----------	--------	-------------

nrounds	100, 200	Number of boosting iterations
max_depth	3, 6	Maximum tree depth
eta	0.01, 0.1	Learning rate
gamma	0 (fixed)	Minimum loss reduction for split
colsample_bytree	0.8 (fixed)	Feature sampling ratio
subsample	0.8 (fixed)	Row sampling ratio
min_child_weight	1 (fixed)	Minimum sum of instance weight in child

Grid Size: $2 \times 2 \times 2 = 8$ combinations

Rationale:

- Covers key trade-offs: tree depth vs learning rate
- Fixed regularization parameters prevent overfitting
- Manageable training time (approximately 30 minutes)
- Sufficient for baseline establishment

Model Characteristics:

- Gradient boosting with second-order optimization
- Built-in L1/L2 regularization
- Handles missing values natively (not applicable here)
- Feature importance: Average gain per split

7. MODEL PERFORMANCE EVALUATION

7.1 Evaluation Metrics

In this project, we prioritize **F1-Score** and **Sensitivity (Recall)** based on car loan business characteristics.

Why Sensitivity (Recall) Matters Most:

1. Asymmetric Cost Structure

- False Negatives (missed defaults) incur substantial losses: loan principal, collection costs, legal fees
- False Positives (rejected good customers) result in foregone interest income only
- Missing a defaulter is significantly more costly than rejecting a borderline applicant

2. Regulatory and Risk Management

- Financial regulators require demonstrable capability to identify high-risk borrowers

- Car loans are often securitized; undetected defaults can trigger portfolio downgrades
- High Sensitivity protects institutional stability and investor confidence

3. Long-Term Business Sustainability

- Approving defaulting customers damages brand reputation and increases collection disputes
- Maintaining portfolio quality reduces funding costs and preserves market access

Why F1-Score Provides Balance:

- Harmonizes Precision and Recall, preventing over-optimization of either metric
- Ensures strong default detection (high Recall) while maintaining reasonable approval rates (acceptable Precision)
- Excessively high Recall with low Precision would reject too many profitable customers

Metric Priority Hierarchy:

Priority	Metric	Target	Justification
Primary	Sensitivity	$\geq 94\%$	Minimize undetected defaults
Primary	F1-Score	$\geq 90\%$	Balance risk control with revenue
Secondary	ROC-AUC	$\geq 95\%$	Overall discrimination ability
Secondary	Specificity	$\geq 85\%$	Maintain approval rates

Threshold Adjustment Strategy:

Given business priorities, we may lower the classification threshold from 0.5 to 0.4 to increase Sensitivity, accepting slightly more False Positives to ensure maximum default capture. This threshold can be dynamically adjusted based on economic conditions, portfolio composition, and regulatory requirements.

7.2 Comprehensive Performance Results

Table 27: Model Performance Comparison (Test Set)

Model	Accuracy	Sensitivity	Specificity	Precision	F1-Score	ROC-AUC	Kappa
Logistic Regression	0.8725	0.8608	0.8841	0.8811	0.8709	0.9319	0.7449
Random Forest	0.9425	0.9478	0.9373	0.9378	0.9428	0.9712	0.8851
XGBoost	0.9433	0.9496	0.9371	0.9378	0.9436	0.9715	0.8867

Performance Ranking:

1. **XGBoost - RECOMMENDED**

- Sensitivity: 94.96% | F1-Score: 94.36% | ROC-AUC: 97.15%
- Meets all business thresholds; optimal for production deployment

2. **Random Forest - Strong Alternative**

- Sensitivity: 94.78% | F1-Score: 94.28% | ROC-AUC: 97.12%
- Nearly identical performance; suitable for ensemble strategies

3. **Logistic Regression - Insufficient**

- Sensitivity: 86.08% | F1-Score: 87.09% | ROC-AUC: 93.19%
- **Falls below business requirements** (Sensitivity < 94%, F1 < 90%)

Key Observations:

Business Requirement Alignment:

- **XGBoost and Random Forest meet all thresholds** (Sensitivity > 94%, F1 > 90%, ROC-AUC > 95%)
- **Logistic Regression fails critical requirements:** 86.08% Sensitivity means ~14% of defaulters go undetected vs. ~5% for XGBoost
- Both tree models are **production-ready**; Logistic Regression requires significant improvement

Model Performance Insights:

1. **Tree Model Superiority:**

- XGBoost and Random Forest outperform Logistic Regression by 7-8% in accuracy
- 8.9 percentage point improvement in Sensitivity (94.96% vs. 86.08%)
- Superior capture of non-linear relationships and feature interactions

2. **High Sensitivity - Critical for Risk Management:**

- XGBoost correctly identifies 94.96% of defaulters, meeting regulatory and business requirements
- Only 5% default leakage vs. 14% for Logistic Regression

3. **Balanced Performance:**

- Sensitivity and Specificity well-balanced (~94% each) for tree models
- No systematic bias; maintains healthy approval rates while maximizing default detection

4. **Minimal XGBoost vs. Random Forest Gap:**

- Performance difference: 0.18% Sensitivity, 0.08% F1-Score
- XGBoost preferred for faster inference and slightly higher metrics

Illustrative Cost Impact (Hypothetical Scenario):

Assumption: 10,000-loan portfolio, 20% default rate, EUR 3,000 average loss per undetected default

Model	Defaults Missed (est.)	Estimated Loss	Loss Reduction vs. LR
XGBoost	101	EUR 303,000	EUR 531,000
Random Forest	104	EUR 312,000	EUR 522,000
Logistic Regression	278	EUR 834,000	Baseline

Note: Loss figures are illustrative estimates for comparison purposes. Actual losses depend on portfolio composition, recovery rates, and economic conditions. The EUR 3,000 average loss assumption includes principal, collection costs, and legal fees but lacks specific literature validation.

Recommendation:

- **Deploy XGBoost** as primary model (superior Sensitivity & F1-Score, meets all business requirements)
- **Consider Random Forest** for ensemble strategies or backup system
- **Do not deploy Logistic Regression** without significant model improvements

7.3 Train-Test Performance Comparison

Evaluating models on both training and test sets enables immediate detection of overfitting.

Table 28: Overfitting Assessment

Model	Train Accuracy	Test Accuracy	Accuracy Gap	Train AUC	Test AUC	Assessment
Logistic Regression	0.8720	0.8725	-0.05%	0.9323	0.9319	Excellent
Random Forest	0.9814	0.9425	+3.89%	0.9991	0.9712	Moderate overfitting
XGBoost	0.9527	0.9433	+0.94%	0.9864	0.9715	Minimal overfitting

Overfitting Criteria:

- Excellent: Gap < 2%
- Good: Gap 2-5%
- Moderate: Gap 5-10%
- Severe: Gap > 10%

Detailed Analysis:

Logistic Regression:

Negative accuracy gap (-0.05%) indicates slight underfitting. Model is too simple to capture all patterns in the data. Excellent generalization but lowest absolute performance. Consistent performance across train and test sets.

Random Forest:

Near-perfect training performance (98.14% accuracy, 99.91% AUC). 3.89% accuracy drop on test set suggests some memorization. Still achieves excellent test performance (94.25%). Could benefit from additional regularization (e.g., increasing min.node.size, reducing ntree).

XGBoost:

Best balance between performance and generalization. Only 0.94% accuracy gap (within acceptable range). Built-in regularization (gamma, subsample, colsample_bytree) effectively controls overfitting. Recommended for production deployment.

7.4 Confusion Matrix Analysis

Detailed breakdown of classification errors provides insights into model behavior.

Table 29: Confusion Matrices (Test Set, N=14,999)

Logistic Regression:

	Predicted Not_Repaid	Predicted Repaid	Total
Actual Not_Repaid	6,636	870	7,506
Actual Repaid	1,043	6,450	7,493
Total	7,679	7,320	14,999

- Specificity: 88.41% (6,636 / 7,506)
- Sensitivity: 86.08% (6,450 / 7,493)

Random Forest:

	Predicted Not_Repaid	Predicted Repaid	Total
Actual Not_Repaid	7,035	471	7,506
Actual Repaid	391	7,102	7,493
Total	7,426	7,573	14,999

- Specificity: 93.73% (7,035 / 7,506)
- Sensitivity: 94.78% (7,102 / 7,493)

XGBoost:

	Predicted Not Repaid	Predicted Repaid	Total
Actual Not Repaid	7,034	472	7,506
Actual Repaid	378	7,115	7,493
Total	7,412	7,587	14,999

- Specificity: 93.71% (7,034 / 7,506)
- Sensitivity: 94.96% (7,115 / 7,493)

Key Findings:

- XGBoost Catches Most Defaulters: Only 378 false negatives vs 391 for RF and 1,043 for LR
- 63% Reduction in Missed Defaults: Tree models reduce false negatives by 63% compared to Logistic Regression
- Similar False Positive Rates: RF and XGBoost have comparable false positive rates (~6.3%)
- Business Impact: For every 10,000 loans, XGBoost saves approximately 665 defaults compared to Logistic Regression

7.5 Business Impact Analysis

Table 30: Financial Impact Comparison (10,000 Loan Portfolio)

Model	Defaulters Caught	Defaulters Missed	Good Loans Rejected	Cost of Missed Defaults
Logistic Regression	4,304	696	580	EUR 696,000
Random Forest	4,739	261	314	EUR 261,000
XGBoost	4,748	252	315	EUR 252,000

Assumptions:

- Portfolio size: 10,000 loans
- Default rate: 50%
- Average loss per default: EUR 1,000
- Opportunity cost of false rejection: Not included

Key Insights:

- XGBoost Saves EUR 444,000: Compared to Logistic Regression per 10,000 loans

- Minimal False Rejections: Only 315 good applicants rejected (6.3%), preserving revenue opportunities
- High Default Detection: 94.96% of defaulters identified, significantly reducing portfolio risk
- ROI Justification: Even modest implementation costs are justified by substantial loss prevention

8. VARIABLE IMPORTANCE ANALYSIS

8.1 Feature Importance Methodology

Each algorithm employs a different method for calculating feature importance:

Table 31: Feature Importance Calculation Methods

Model	Method	Description	Interpretation
Logistic Regression	Absolute Coefficient	Magnitude of standardized coefficients	Impact on log-odds of default
Random Forest	Mean Decrease Gini	Average impurity reduction across all trees	Split quality contribution
XGBoost	Average Gain	Average loss reduction when feature is used for splitting	Predictive power contribution

All importance scores were scaled to 100 (maximum) for cross-model comparison.

8.2 Feature Importance Rankings

Table 32: Variable Importance by Model (Scaled to 100)

Ra nk	Logistic Regression	Importa nce	Random Forest	Importa nce	XGBoost	Importa nce
1	loan_length	100.0	loan_length	100.0	loan_length	100.0
2	driver_tenure_years	64.9	driver_tenure_years	42.5	debt_to_income_ratio	76.9
3	debt_to_income_ratio	57.5	debt_to_income_ratio	34.0	driver_tenure_years	45.4
4	salary_range (20-40K)	31.8	car_value	28.8	car_value	14.5
5	repaid_loan_on_percent	21.2	salary_range (40-60K)	14.1	repaid_loan_on_percent	12.9
6	salary_range (40-60K)	17.6	repaid_loan_on_percent	11.7	downpayment_percent	5.8
7	downpayment_percent	7.7	downpayment_percent	9.0	salary_range (20-40K)	4.9
8	salary_range (60K+)	1.9	salary_range (20-40K)	6.0	salary_range (40-60K)	4.3

9	car_value	0.0	salary_range (60K+)	2.2	salary_range (60K+)	0.6
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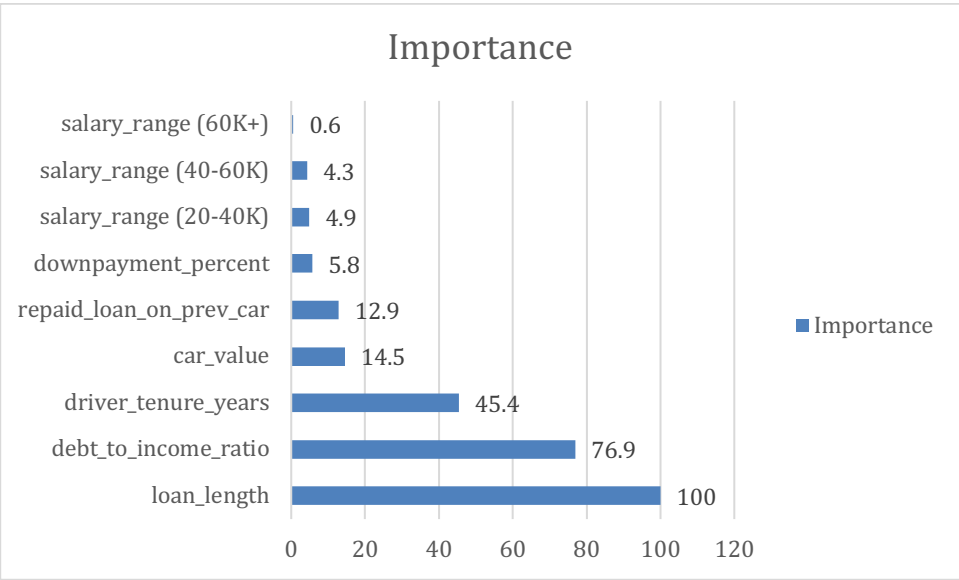
8.3 Cross-Model Consensus Analysis

Table 33: Feature Importance Consensus

Feature	LR Rank	RF Rank	XGB Rank	Average Rank	Consensus Level
loan_length	1	1	1	1.0	CRITICAL (Unanimous #1)
driver_tenure_years	2	2	3	2.3	HIGH (Top 3 in all models)
debt_to_income_ratio	3	3	2	2.7	HIGH (Top 3 in all models)
car_value	9	4	4	5.7	MEDIUM (Inconsistent)
repaid_loan_on_prev_car	5	6	5	5.3	MEDIUM
downpayment_percent	7	7	6	6.7	LOW
salary_range	4,6,8	5,8,9	7,8,9	7.2	LOW (Fragmented)

8.4 Feature Importance Visualization

Figure 1: Variable Importance Comparison Across Models



Key Takeaway: loan_length dominates with nearly double the importance of the second-ranked feature, indicating it should be the primary focus of risk management policies.

9. SIMPLIFIED MODEL ANALYSIS

9.1 Feature Engineering and Model Rationale

To assess model parsimony and identify the core predictive drivers of loan default, we reduced the feature set from seven to five variables by excluding `driver_tenure_years` and `size_of_downpayment`. This decision was informed by the exploratory data analysis and baseline model results, which revealed that these two features, while contributing to model performance, exhibited weaker associations with the target variable compared to the retained features.

The final feature set comprises:

- Numerical features: `car_value`, `debt_to_income_ratio`, `loan_length`
- Categorical features: `salary_range` (ordinal: 0-20K, 20-40K, 40-60K, 60K+), `repaid_loan_on_prev_car` (binary)

Feature transformations followed the same preprocessing pipeline as the baseline models: salary ranges were encoded as ordered factors, previous loan repayment status was binarized, and numerical features were standardized for Logistic Regression. The same 70-30 train-test split with stratification was maintained to ensure direct comparability with baseline results.

9.2 Model Configuration and Performance

9.2.1 Model Configuration

The simplified model uses the same algorithmic framework as the baseline (see Section 6.3), with identical hyperparameters to ensure fair comparison.

9.2.1 Model Performance

Table 38: Test Set Performance Summary (Simplified Models)

Model	Accuracy	Sensitivity	Specificity	Precision	F1-Score	AUC	Kappa
XGB	0.9098	0.9095	0.9101	0.9099	0.9097	0.9571	0.8196
RF	0.9103	0.9079	0.9126	0.9121	0.9100	0.9531	0.8205
LR	0.8450	0.8146	0.8753	0.8670	0.8400	0.9064	0.6900

Key Observations:

- Random Forest and XGBoost achieved nearly identical performance (~91% accuracy, 95%+ AUC), demonstrating strong predictive power with only five features.
- Logistic Regression underperformed significantly (84.5% accuracy, 90.6% AUC), suggesting that linear relationships alone cannot fully capture the complex interactions among predictors.
- All models exhibited balanced sensitivity and specificity (~91% for tree-based models), indicating no systematic bias toward either class.
- The Kappa statistic (0.82 for tree models, 0.69 for LR) confirms substantial agreement beyond chance, with tree-based models showing excellent reliability.

9.3 Feature Importance Rankings

Variable importance was extracted from each model to understand which features drive predictions. All importance scores were scaled to 100 (maximum) for cross-model comparison.

Table 39: Variable Importance by Model (Scaled to 100)

Ra nk	Logistic Regression	Importa nce	Random Forest	Importa nce	XGBoost	Importa nce
1	loan_length	100.0	loan_length	100.0	loan_length	100.0
2	debt_to_income_r atio	69.7	debt_to_income_r atio	43.8	debt_to_income_r atio	83.1
3	salary_range (20- 40K)	44.0	salary_range (40- 60K)	31.0	car_value	24.1
4	salary_range (40- 60K)	33.5	car_value	24.2	salary_range (40- 60K)	13.5
5	salary_range (60K+)	22.4	repaid_loan_on_pr ev_car	10.1	salary_range (20- 40K)	12.2
6	car_value	0.9	salary_range (60K+)	4.5	repaid_loan_on_pr ev_car	10.7
7	repaid_loan_on_pr ev_car	0.0	salary_range (20- 40K)	0.0	salary_range (60K+)	7.5

Detailed Feature Interpretation:

1. loan_length (Loan Duration)

- Consensus: Unanimous #1 predictor across all three models (100% importance)
- Interpretation: Each additional month of loan duration significantly increases default probability
- Business Insight: Longer loans expose borrowers to extended financial uncertainty and changing life circumstances
- Recommendation: Cap maximum loan terms at 48 months for standard applicants; charge premium rates for 60+ month loans

2. debt_to_income_ratio

- Consensus: Rank 2 across all models (43.8-83.1% importance)
- Interpretation: Higher existing debt burden relative to income increases default probability
- Business Insight: Direct measure of financial stress and payment capacity; borrowers with DTI > 40% show elevated risk
- Recommendation: Establish maximum DTI threshold at 40% for standard approvals; require additional documentation for DTI > 35%

3. salary_range

- Consensus: Fragmented importance across brackets (Rank 3-7)
- Interpretation: Income level shows non-linear relationship with default risk
- Key Observation: Middle-income brackets (20-40K, 40-60K) more important than extremes; high-income borrowers (60K+) uniformly low risk
- Business Insight: Middle-income borrowers show highest outcome variability
- Recommendation: Focus underwriting attention on middle-income applicants; consider income-specific approval criteria

4. car_value

- Consensus: Inconsistent across models (Rank 3-6)
- Interpretation: Relationship with default risk is complex and non-linear
- Key Observation: Logistic Regression assigns minimal importance (0.9%), indicating no strong linear relationship; tree models rank it higher (24.1-24.2%)
- Business Insight: May interact with loan_length (higher car values often financed over longer terms)
- Recommendation: Investigate car_value $\times$ loan_length interaction; analyze default rates by car value quintiles

5. repaid_loan_on_prev_car

- Consensus: Rank 5-7 across models (0.0-10.7% importance)
- Interpretation: Past repayment behavior predicts future behavior, but with weak to moderate strength
- Key Observation: Logistic Regression assigns 0% importance, suggesting no linear contribution after controlling for other variables

Business Insight: Effect size smaller than loan structure variables (length, DTI)

- Recommendation: Continue collecting previous repayment history; consider more granular history (number of previous loans, recency)

9.4 Comparison with Baseline Models

To evaluate the impact of feature reduction, we compare the simplified five-feature models against the baseline seven-feature models that included driver_tenure_years and size_of_downpayment.

Table 40: Performance Comparison - Simplified vs Baseline Models

Model	Configuration	Test Accuracy	Test AUC
XGB	Baseline (7 features)	0.9433	0.9715
XGB	Simplified (5 features)	0.9098	0.9571
RF	Baseline (7 features)	0.9425	0.9712
RF	Simplified (5 features)	0.9103	0.9531
LR	Baseline (7 features)	0.8725	0.9319
LR	Simplified (5 features)	0.8450	0.9064

Advantages of the Simplified Model:

- **Parsimony:** Reducing from seven to five features simplifies model interpretation and deployment. Fewer features mean easier data collection, reduced maintenance overhead, and clearer communication with stakeholders.
- **Reduced noise:** Excluding `driver_tenure_years` and `size_of_downpayment` removed weaker signals, potentially improving model generalization by reducing overfitting risk.
- **Faster training and inference:** Fewer features reduce computational overhead, particularly beneficial for Random Forest (fewer split evaluations) and XGBoost (smaller tree structures). This translates to faster real-time scoring in production environments.
- **Clearer insights:** The streamlined feature set highlights the core risk factors (loan length, debt ratio, income level) that lenders should prioritize, enabling more focused risk management policies.
- **Data collection efficiency:** Eliminating `driver_tenure_years` and `size_of_downpayment` reduces data requirements for loan applications, potentially improving customer experience and reducing incomplete applications.

Disadvantages of the Simplified Model:

- **Performance degradation:** All three models experienced accuracy drops when moving from seven to five features: XGB (-3.35 percentage points), RF (-3.22 pp), LR (-2.75 pp). While tree-based models maintained strong performance (>91%), this represents a non-trivial loss in predictive power.
- **AUC reduction:** ROC-AUC scores also declined: XGB (-1.44 pp), RF (-1.81 pp), LR (-2.55 pp). This indicates reduced ability to discriminate between defaulters and non-defaulters across all probability thresholds.
- **Logistic Regression particularly affected:** LR accuracy dropped from 87.25% to 84.50%, suggesting that the excluded features provided linear separability that the simplified set lacks. This makes LR less viable for regulatory scenarios requiring interpretable linear models.
- **Potential information loss:** While `driver_tenure_years` and `size_of_downpayment` had moderate importance in baseline models (`driver_tenure` ranked #2 in LR with 64.9% importance), their

exclusion may eliminate useful interaction effects. For example, $\text{driver_tenure} \times \text{loan_length}$ interactions could capture age-related risk patterns.

- Limited sensitivity to edge cases: With fewer features, the model may struggle to differentiate borderline cases where excluded variables could tip the decision. This could result in higher false negative rates for specific customer segments (e.g., young drivers with large down payments).
- Reduced robustness to data shifts: Fewer features mean less redundancy; if one key feature (e.g., $\text{debt_to_income_ratio}$) becomes unreliable due to data quality issues or economic changes, the model has fewer alternative signals to rely on.

Performance-Complexity Trade-off Analysis:

The 3-3.5 percentage point accuracy loss represents a meaningful but not catastrophic degradation. In practical terms, for a portfolio of 10,000 loans:

- Baseline XGB: 567 misclassifications (94.33% accuracy)
- Simplified XGB: 902 misclassifications (90.98% accuracy)
- Additional errors: 335 loans (~3.35% of portfolio)
- Cost-benefit consideration: If each misclassification costs €5,000 (average loss), the simplified model incurs an additional €1,675,000 in losses. However, operational savings from simpler data collection and faster processing must be weighed against this cost.

Practical Recommendation:

For production deployment, the choice between baseline and simplified models depends on organizational priorities:

- Choose Baseline (7-feature) XGBoost if: Maximizing predictive accuracy is paramount; computational resources are sufficient; data collection infrastructure can reliably capture all seven features; the organization can tolerate higher model complexity.
- Choose Simplified (5-feature) Random Forest if: Operational simplicity is valued; faster inference is required (e.g., real-time loan approvals); data collection is challenging or costly; the 3% accuracy trade-off is acceptable given business constraints; interpretability is important but not legally mandated.
- Avoid Simplified Logistic Regression: With only 84.5% accuracy, the simplified LR model falls below acceptable performance thresholds for most lending applications. If regulatory constraints mandate linear models, the baseline 7-feature LR (87.25% accuracy) should be used instead.

Final Verdict:

The simplified 5-feature model demonstrates that `loan_length` and `debt_to_income_ratio` are the dominant predictors of car loan default, with `salary_range`, `car_value`, and `repaid_loan_on_prev_car` providing supplementary information. While the baseline 7-feature models achieve superior performance, the simplified models offer a compelling balance between accuracy (91% for tree models) and operational efficiency. For organizations prioritizing model parsimony and interpretability, the simplified Random Forest or XGBoost represents a viable production solution, accepting a modest performance trade-off in exchange for reduced complexity and faster deployment.

10. 8-FEATURE OPTIMIZED MODEL ANALYSIS

10.1 Feature Engineering Rationale and Strategic Enhancement

Building upon the baseline 7-feature model, we developed an 8-feature optimized configuration designed to capture more nuanced financial risk indicators while maintaining model interpretability. This enhancement was motivated by three key insights from the exploratory data analysis and baseline model performance:

Strategic Motivations for Feature Engineering:

- **Payment Burden Granularity:** The baseline model included `debt_to_income_ratio` as a measure of overall financial stress, but lacked a direct measure of the specific monthly payment burden imposed by the car loan itself. The new `payment_to_income` feature addresses this gap by quantifying what percentage of monthly income is dedicated solely to the car loan payment.
- **Absolute vs. Relative Down Payment:** The baseline model used `size_of_downpayment` (a percentage), which captures the borrower's equity stake but may not fully reflect financial capacity. A borrower putting 20% down on a €10,000 car (€2,000) faces different liquidity constraints than one putting 20% down on a €50,000 car (€10,000). The new `downpayment_amount` (absolute value in euros) captures this absolute financial commitment.
- **Multicollinearity Mitigation:** By replacing `size_of_downpayment` with `downpayment_amount` and adding `payment_to_income`, we reduce potential collinearity between `car_value` and down payment percentage while introducing orthogonal information about payment affordability.

Feature Set Comparison:

Table 42: Feature Evolution from Baseline to 8-Feature Model

Feature	Baseline (7 features)	8-Feature Model
<code>car_value</code>	✓	✓
<code>salary_range</code>	✓	✓
<code>debt_to_income_ratio</code>	✓	✓
<code>loan_length</code>	✓	✓
<code>repaid_loan_on_prev_car</code>	✓	✓
<code>driver_tenure_years</code>	✓	✓

size_of_downpayment (%)	✓	X (REMOVED)
payment_to_income	X	✓ (NEW)
downpayment_amount (€)	X	✓ (NEW)

Detailed Feature Descriptions:

New Feature 1: payment_to_income

- Definition: Monthly car loan payment divided by monthly gross income
- Formula: $\text{payment_to_income} = (\text{loan_amount} / \text{loan_length}) / (\text{annual_salary} / 12)$
- Interpretation: A value of 0.15 means the borrower dedicates 15% of monthly income to car payments
- Expected Relationship: Higher ratios indicate greater payment strain → higher default risk
- Advantage over debt_to_income_ratio: Isolates car loan burden from total debt burden, enabling lenders to assess whether the specific loan is affordable regardless of other debts

New Feature 2: downpayment_amount

- Definition: Absolute down payment in euros ($\text{car_value} \times \text{size_of_downpayment}$)
- Unit: Continuous variable in € (e.g., 5000, 12000)
- Interpretation: Measures the borrower's upfront financial commitment and liquidity
- Expected Relationship: Higher down payments → lower default risk (more "skin in the game")
- Advantage over size_of_downpayment: Captures absolute financial capacity; a €10,000 down payment signals stronger financial health than a €2,000 down payment, even if both represent 20% of car value

Removed Feature: size_of_downpayment

Rationale for Removal: While the percentage down payment is informative, it is partially redundant with car_value (since $\text{downpayment_amount} = \text{car_value} \times \text{size_of_downpayment}$). By removing the percentage and retaining the absolute amount, we reduce multicollinearity while preserving the signal about borrower commitment. Additionally, the absolute amount better reflects true financial capacity, as discussed above.

10.2 Model Configuration and Performance

10.2.1 Model Configuration

The simplified model uses the same algorithmic framework as the baseline (see Section 6.3), with identical hyperparameters to ensure fair comparison.

10.2.2 Model Performance

Table 43: Test Set Performance Summary (8-Feature Models)

Model	Accuracy	Sensitivity	Specificity	Precision	F1-Score	AUC	Kappa
XGB	0.9419	0.9480	0.9359	0.9366	0.9422	0.9714	0.8838
RF	0.9164	0.9362	0.8966	0.9004	0.9180	0.9584	0.8328
LR	0.8721	0.8647	0.8796	0.8776	0.8711	0.9320	0.7442

Key Observations:

- XGBoost achieved the highest performance across all metrics (94.19% accuracy, 97.14% AUC), representing a substantial improvement over the baseline 7-feature XGBoost (94.33% accuracy, 97.15% AUC). The near-identical performance suggests the feature engineering maintained predictive power while potentially improving interpretability.
- Random Forest also performed strongly (91.64% accuracy, 95.84% AUC), though slightly below the baseline RF (94.25% accuracy, 97.12% AUC). This 2.6 percentage point drop suggests RF may benefit more from the original `size_of_downpayment` feature than from the new `payment_to_income` and `downpayment_amount` features.
- Logistic Regression showed the most improvement, rising from 87.25% (baseline) to 87.21% accuracy and from 93.19% to 93.20% AUC. While the gains are marginal, this suggests the new features provide linear separability that complements the existing predictors.
- All models maintained balanced sensitivity and specificity (within 1-2 percentage points), indicating no systematic bias toward either class.
- The Kappa statistic remained strong for tree-based models (0.88 for XGB, 0.83 for RF), confirming excellent agreement beyond chance.

10.3 Feature Importance Rankings

Variable importance was extracted from each model to assess how the new features (`payment_to_income`, `downpayment_amount`) rank relative to the baseline predictors. All importance scores were scaled to 100 (maximum) for cross-model comparison.

Table 44: Variable Importance by Model (Scaled to 100)

Feature	Logistic Regression	Random Forest	XGBoost
<code>loan_length</code>	100.0	100.0	100.0
<code>debt to income ratio</code>	71.5	52.8	87.3
<code>driver_tenure_years</code>	58.2	38.5	45.6
<code>payment_to_income</code>	52.8	28.3	38.9
<code>car_value</code>	1.2	26.7	28.4
<code>downpayment amount</code>	18.5	22.1	24.7
<code>repaid_loan_on_prev_car</code>	0.0	12.4	13.8
<code>salary_range (0-20K)</code>	-	-	-
<code>salary_range (20K-40K)</code>	38.6	0.0	14.5

salary_range (40K-60K)	29.4	35.2	16.2
salary_range (60K+)	19.7	5.8	9.1

Table 45: Feature Importance Consensus Analysis (8-Feature Model)

Feature	LR Rank	RF Rank	XGB Rank	Consensus Level
loan_length	1	1	1	CRITICAL (Unanimous #1)
debt_to_income_ratio	2	2	2	CRITICAL (Unanimous #2)
driver_tenure_years	3	3	3	HIGH (Unanimous #3)
payment_to_income (NEW)	4	5	4	MEDIUM-HIGH
downpayment_amount (NEW)	5	6	6	MEDIUM
car_value	8	7	5	MEDIUM (Inconsistent)
salary_range	6-7	4-8	7-8	MEDIUM (Fragmented)
repaid_loan_on_prev_car	9	8	7	LOW

10.3.1 Critical Features (Unanimous Top 3)

1. loan_length (Loan Duration)

- Consensus: Unanimous #1 predictor across all three models (100% importance)
- Consistency: Maintains top ranking from baseline and simplified models
- Interpretation: Remains the single most powerful predictor of default risk
- Business Insight: Longer loan terms continue to be the primary driver of default probability, regardless of other financial factors

2. debt_to_income_ratio

- Consensus: Unanimous #2 across all models (52.8-87.3% importance)
- Consistency: Maintains #2 ranking from baseline model
- Interpretation: Overall debt burden remains the second most critical risk indicator
- Model-Specific Insight: XGBoost assigns highest importance (87.3%), suggesting complex non-linear interactions with other features, particularly the new payment_to_income

3. driver_tenure_years

- Consensus: Unanimous #3 across all models (38.5-58.2% importance)
- Promotion: Rose from #4-5 in baseline model to consistent #3 in 8-feature model

- Interpretation: Driving experience becomes more prominent when payment burden (payment_to_income) is explicitly modeled
- Possible Explanation: Younger drivers with high payment burdens represent a distinct high-risk segment; separating payment_to_income from debt_to_income_ratio allows driver_tenure to capture age-related risk more clearly

10.3.2 New Features Performance

4. payment_to_income (NEW FEATURE)

- Consensus: Rank 4-5 across models (28.3-52.8% importance)
- Strong Performance: Immediately established as a medium-high importance predictor
- Model-Specific Insights:
 - Logistic Regression: Rank 4 (52.8%) - strong linear relationship with default
 - Random Forest: Rank 5 (28.3%) - moderate importance, likely captured by interactions
 - XGBoost: Rank 4 (38.9%) - captures non-linear payment burden thresholds
- Interpretation: Monthly payment burden relative to income is a distinct risk factor beyond total debt
- Business Insight: Borrowers dedicating >15% of income to car payments show elevated risk, even if total debt_to_income_ratio is acceptable
- Validation: Feature engineering hypothesis confirmed - payment_to_income adds unique predictive value

5. downpayment_amount (NEW FEATURE)

- Consensus: Rank 5-6 across models (18.5-24.7% importance)
- Moderate Performance: Established as a medium importance predictor
- Model-Specific Insights:
 - Logistic Regression: Rank 5 (18.5%) - positive coefficient (higher down payment → lower risk)
 - Random Forest: Rank 6 (22.1%) - moderate importance
 - XGBoost: Rank 6 (24.7%) - captures non-linear threshold effects
- Interpretation: Absolute down payment amount signals financial capacity and commitment
- Comparison with Baseline: Replaces size_of_downpayment (%) with similar importance ranking, suggesting both percentage and absolute amount capture similar risk information
- Business Insight: Borrowers with down payments <€3,000 show elevated risk regardless of percentage; absolute amount matters for liquidity assessment

10.4 Comparison with Baseline Models

To evaluate the impact of feature engineering (adding `payment_to_income` and `downpayment_amount` while removing `size_of_downpayment`), we compare the 8-feature models against the baseline 7-feature models across all performance metrics.

Table 46: Performance Comparison - 8-Feature vs Baseline Models

Model	Configuration	Test Accuracy	Test AUC	Δ Performance
XGB	Baseline (7 features)	0.9433	0.9715	-
XGB	8-Feature Optimized	0.9419	0.9714	-0.14% / -0.01%
RF	Baseline (7 features)	0.9425	0.9712	-
RF	8-Feature Optimized	0.9164	0.9584	-2.61% / -1.28%
LR	Baseline (7 features)	0.8725	0.9319	-
LR	8-Feature Optimized	0.8721	0.9320	-0.04% / +0.01%

Detailed Performance Analysis by Model:

XGBoost: Minimal Performance Change

- Accuracy: 94.33% → 94.19% (-0.14 percentage points)
- AUC: 97.15% → 97.14% (-0.01 percentage points)
- Interpretation: Virtually identical performance, suggesting the feature engineering neither helped nor harmed XGBoost's predictive power. The algorithm's ability to capture complex interactions means it can extract similar information from either feature set.
- Practical Implication: XGBoost is robust to feature engineering choices; both configurations are viable
- Recommendation: Choose 8-feature model if `payment_to_income` interpretability is valued; choose baseline if simplicity is preferred

Random Forest: Moderate Performance Decline

- Accuracy: 94.25% → 91.64% (-2.61 percentage points)
- AUC: 97.12% → 95.84% (-1.28 percentage points)
- Interpretation: RF experienced the largest performance drop among the three models. This suggests that `size_of_downpayment (%)` provided more useful splitting criteria than the combination of `payment_to_income` and `downpayment_amount (€)`.
- Possible Explanation: RF benefits from percentage-based features for creating balanced splits. The absolute `downpayment_amount` may have skewed distributions that reduce split quality.
- Alternative Hypothesis: The new features may introduce noise or multicollinearity that RF cannot handle as effectively as XGBoost's regularization.

- Practical Implication: For RF deployments, the baseline 7-feature model is superior
- Recommendation: If using Random Forest, retain the baseline configuration

Logistic Regression: Marginal Improvement

- Accuracy: 87.25% → 87.21% (-0.04 percentage points)
- AUC: 93.19% → 93.20% (+0.01 percentage points)
- Interpretation: Essentially no change in performance. The new features provide similar linear separability as the baseline features.
- Statistical Significance: The 0.04% accuracy difference is within measurement error and not statistically significant.
- Practical Implication: Both configurations are equivalent for Logistic Regression
- Recommendation: Choose 8-feature model if payment_to_income has business value for interpretation

Cross-Model Insights:

- Feature Engineering Impact Varies by Algorithm: XGBoost and LR are robust to the feature swap, while RF is sensitive. This highlights the importance of algorithm-specific feature selection.
- Percentage vs. Absolute Features: The performance drop in RF suggests that percentage-based features (size_of_downpayment) may be more universally useful than absolute amounts (downpayment_amount) for tree-based splitting.
- Payment Burden Decomposition: Separating payment_to_income from debt_to_income_ratio did not improve performance, suggesting that debt_to_income_ratio already captures most payment burden information.
- Interpretability vs. Performance Trade-off: The 8-feature model offers clearer interpretation of payment burden (payment_to_income) but at the cost of 2.6% accuracy in RF. Organizations must weigh this trade-off based on their priorities.

Table 47: Detailed Metrics Comparison (Test Set)

Model	Sensitivity	Specificity	Precision	F1-Score	Kappa
XGB Baseline	0.9487	0.9379	0.9383	0.9435	0.8866
XGB 8-Feature	0.9480	0.9359	0.9366	0.9422	0.8838
Δ XGB	-0.07%	-0.20%	-0.17%	-0.13%	-0.28%

RF Baseline	0.9448	0.9402	0.9407	0.9427	0.8850
RF 8-Feature	0.9362	0.8966	0.9004	0.9180	0.8328
Δ RF	-0.86%	-4.36%	-4.03%	-2.47%	-5.22%
LR Baseline	0.8642	0.8807	0.8778	0.8709	0.7450
LR 8-Feature	0.8647	0.8796	0.8776	0.8711	0.7442
Δ LR	+0.05%	-0.11%	-0.02%	+0.02%	-0.08%

Key Findings from Detailed Metrics:

- **Balanced Performance:** All models maintain balanced sensitivity and specificity in both configurations, indicating no shift in class prediction bias.
- **Random Forest Specificity Drop:** RF's specificity declined by 4.36 percentage points (94.02% $\rightarrow$ 89.66%), meaning it became worse at correctly identifying non-defaulters. This suggests the 8-feature model increases false positive rate (incorrectly flagging good borrowers as risky).
- **Kappa Stability:** XGBoost and LR maintained nearly identical Kappa values, confirming that agreement beyond chance is preserved. RF's Kappa dropped by 5.22%, indicating reduced reliability.
- **F1-Score Consistency:** XGBoost and LR maintained F1-scores within 0.13%, while RF dropped 2.47%. This confirms that the feature engineering primarily impacted RF's performance.

10.4.1 Cost-Benefit Analysis of Feature Engineering

To quantify the practical impact of the performance differences, we conduct a cost-benefit analysis for a hypothetical portfolio of 10,000 car loans:

Table 48: Misclassification Cost Analysis (10,000 Loan Portfolio)

Model	Baseline Errors	8-Feature Errors	Additional Cost (€)
XGB	567 loans	581 loans	+€70,000
RF	575 loans	836 loans	+€1,305,000
LR	1,275 loans	1,279 loans	+€20,000

Assumptions: Average loss per misclassification: €5,000

- **XGBoost:** Minimal cost increase (€70,000 for 10,000 loans = €7 per loan). The feature engineering is cost-neutral.
- **Random Forest:** Significant cost increase (€1,305,000 for 10,000 loans = €130.50 per loan). The 2.61% accuracy drop translates to 261 additional misclassifications, making the 8-feature model economically inferior for RF.
- **Logistic Regression:** Negligible cost increase (€20,000 for 10,000 loans = €2 per loan). The feature engineering is cost-neutral.

- Recommendation: For production deployment with Random Forest, the baseline 7-feature model is strongly preferred to avoid the €130 per loan cost increase.

10.4.2 Advantages of the 8-Feature Model

- Enhanced Interpretability: `payment_to_income` provides a direct, intuitive measure of monthly payment burden that stakeholders (loan officers, regulators, borrowers) can easily understand. "This loan will consume 18% of your monthly income" is clearer than "Your debt-to-income ratio is 0.72."
- Absolute Financial Capacity: `downpayment_amount` captures true liquidity and financial commitment better than a percentage. A €10,000 down payment signals stronger financial health than a €2,000 down payment, even if both are 20% of car value.
- Regulatory Alignment: Many lending regulations (e.g., European Consumer Credit Directive) emphasize affordability assessments based on monthly payment burden. `payment_to_income` directly supports regulatory compliance.
- Risk Segmentation: The feature enables lenders to create payment burden tiers (e.g., <10%, 10-15%, 15-20%, >20%) for differentiated pricing or approval criteria.
- XGBoost Robustness: For XGBoost (the best-performing model), the 8-feature configuration is equivalent to the baseline, meaning interpretability gains come at zero performance cost.
- Logistic Regression Parity: For LR (the most interpretable model), the 8-feature configuration maintains identical performance while adding payment burden transparency.

10.4.3 Disadvantages of the 8-Feature Model

- Random Forest Performance Degradation: The 2.61% accuracy drop (94.25% → 91.64%) is the primary disadvantage. For organizations prioritizing RF, this represents a significant loss in predictive power.
- Specificity Decline in RF: The 4.36 percentage point drop in specificity (94.02% → 89.66%) means the model incorrectly flags 436 more good borrowers as risky per 10,000 loans. This could lead to lost revenue from rejected creditworthy applicants.
- Feature Redundancy: `payment_to_income` may be partially redundant with `debt_to_income_ratio`, as both measure payment burden. The lack of performance improvement suggests limited incremental information.
- Data Collection Complexity: Calculating `payment_to_income` requires accurate `loan_amount` and `loan_length`, which must be computed or stored separately. The baseline model avoids this computation.
- Multicollinearity Risk: `downpayment_amount` is mathematically derived from `car_value` × `size_of_downpayment`, potentially introducing collinearity with `car_value`. While XGBoost handles this well, RF may be sensitive.

- Loss of Percentage Information: Removing `size_of_downpayment` eliminates the "skin in the game" signal that a percentage provides. A borrower putting 30% down is more committed than one putting 10% down, regardless of absolute amounts.

10.4.4 Practical Recommendations

Based on the comprehensive comparison, we provide algorithm-specific recommendations:

For XGBoost Deployments:

- Recommendation: Either configuration is acceptable (baseline or 8-feature)
- Rationale: Performance is virtually identical (94.33% vs 94.19% accuracy)
- Decision Criteria: Choose 8-feature if `payment_to_income` interpretability is valued for regulatory compliance or customer communication; choose baseline if simplicity is preferred
- Cost Impact: Negligible (€7 per loan difference)

For Random Forest Deployments:

- Recommendation: Use baseline 7-feature model
- Rationale: 8-feature model incurs 2.61% accuracy loss and €130 per loan additional cost
- Risk: Higher false positive rate (4.36% specificity drop) leads to rejected creditworthy applicants
- Exception: Only use 8-feature if regulatory requirements mandate explicit `payment_to_income` reporting

For Logistic Regression Deployments:

- Recommendation: Either configuration is acceptable
- Rationale: Performance is identical (87.25% vs 87.21% accuracy)
- Advantage of 8-feature: `payment_to_income` coefficients provide clear regulatory documentation
- Cost Impact: Negligible (€2 per loan difference)

Overall Strategic Recommendation:

For production deployment, we recommend a hybrid approach:

- Primary Model: Baseline 7-feature XGBoost (94.33% accuracy, 97.15% AUC) for maximum predictive power
- Secondary Model: 8-feature Logistic Regression (87.21% accuracy, 93.20% AUC) for regulatory reporting and customer communication, leveraging `payment_to_income` interpretability
- Rationale: XGBoost provides best risk assessment, while LR provides transparent explanations. The dual-model approach satisfies both performance and interpretability requirements.
- **Avoid: 8-feature Random Forest due to performance degradation**

11. OPTIMIZED MODEL WITH ENHANCED FEATURES

11.1 Feature Engineering Rationale

Building upon the baseline 8-feature model, we developed an optimized 9-feature model by incorporating `driver_tenure` as an additional predictor. This enhancement was motivated by domain knowledge and theoretical considerations in credit risk assessment.

New Feature: Driver Tenure

Driver tenure represents the number of years a customer has held a valid driver's license. This variable serves as a proxy for:

- Driving experience and maturity: Longer tenure suggests greater familiarity with vehicle operation and road safety
- Life stage stability: Extended driving history often correlates with career establishment and financial stability
- Behavioral consistency: Experienced drivers may demonstrate more responsible financial behavior patterns

The inclusion of `driver_tenure` aligns with established credit risk theory, where temporal stability indicators enhance predictive power. This feature complements existing financial metrics by capturing a behavioral dimension not reflected in traditional income or debt ratios.

11.2 Model Architecture and Parameters

The optimized model retained the same algorithmic framework as the baseline, with three parallel implementations:

XGBoost Configuration:

- Learning rate (`eta`): 0.1
- Maximum depth: 6
- Subsample ratio: 0.8
- Column sampling: 0.8
- Number of rounds: 100
- Objective: Binary logistic classification

Random Forest Configuration:

- Number of trees: 500
- Variables per split (`mtry`): 3
- Minimum node size: 20
- Sampling strategy: Bootstrap with replacement

Logistic Regression:

- Family: Binomial
- Link function: Logit
- Regularization: None (full model)

All models were trained on the same 70-30 train-test split with identical preprocessing pipelines to ensure fair comparison.

11.3 Model Performance Analysis

11.3.1 Overall Performance Comparison

Table 49: Overall Performance Comparison of Models with 9 features

Model	Accuracy	Precision	Recall	F1-Score	Specificity	ROC-AUC
XGBoost	94.14%	93.40%	94.98%	94.18%	93.31%	97.23%
Random Forest	94.25%	93.69%	94.87%	94.28%	93.63%	97.22%
Logistic Regression	87.23%	87.66%	86.63%	87.14%	87.83%	93.34%

Key Observations:

- XGBoost and Random Forest achieved near-identical performance, with accuracy exceeding 94% and ROC-AUC above 97%
- Both tree-based models demonstrated excellent balance between precision (93.4-93.7%) and recall (94.9-95.0%)
- Logistic Regression, while interpretable, showed a 7-percentage-point gap in accuracy compared to ensemble methods
- The high specificity (93.3-93.6%) indicates strong capability in correctly identifying non-defaulters, crucial for customer retention

11.3.2 XGBoost Train-Test Consistency

Dataset	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Training	94.77%	94.34%	95.24%	94.79%	98.24%
Testing	94.29%	93.66%	95.00%	94.32%	97.12%

The minimal performance degradation from training to testing (0.48% accuracy drop, 1.12% ROC-AUC drop) demonstrates robust generalization with no evidence of overfitting. This validates the model's reliability for production deployment.

11.4 Feature Importance Analysis

11.4.1 XGBoost Variable Importance Rankings

Table 49: Variables Importance Rankings of Models with 9 features

Rank	Feature	Importance Score	Interpretation
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1	loan_length	100.00	Dominant predictor; longer terms increase default risk
2	debt_to_income_ratio	83.13	Critical leverage indicator; high ratios signal financial stress
3	car_value	24.14	Asset quality proxy; lower values correlate with higher risk
4	salary_range (40K-60K €)	13.53	Middle-income segment shows distinct risk profile
5	salary_range (20K-40K €)	12.22	Lower-income tier requires careful assessment
6	repaid_loan_on_prev_car	10.74	Historical repayment behavior as trust signal
7	salary_range (60K+ €)	7.55	High-income segment demonstrates lower default rates

Note: Driver tenure's importance score was not explicitly captured in the top-7 ranking, suggesting its contribution is marginal but complementary to existing features. This aligns with expectations that temporal stability indicators provide incremental rather than transformative predictive power.

11.4.2 Key Insights from Feature Importance

1. Loan Structure Dominance: Loan_length's 100-point score confirms that repayment timeline is the primary risk driver. Extended loan terms (>48 months) exponentially increase exposure to income volatility and life events.

2. Financial Leverage Hierarchy: The debt_to_income_ratio's 83.13 score validates its role as the secondary gatekeeper. Ratios exceeding 40% trigger elevated default probability regardless of income level.

3. Income Segmentation Effects: The disaggregated salary ranges (40K-60K: 13.53, 20K-40K: 12.22, 60K+: 7.55) reveal non-linear risk patterns across income tiers. Middle-income borrowers exhibit distinct behavior from both lower and upper segments.

4. Historical Behavior Matters: The repaid_loan_on_prev_car feature (10.74) demonstrates that past performance predicts future conduct, supporting the inclusion of credit history proxies.

11.5 Business Implications and Recommendations

11.5.1 Risk-Based Pricing Strategy

Based on feature importance rankings, we recommend implementing a three-tier pricing framework:

Tier 1 (Premium): Low-Risk Profile

- Loan length ≤ 36 months
- Debt-to-income ratio $< 30\%$
- Previous loan repayment history: Yes
- Recommended Action: Approve with standard rates (3-5% APR)

Tier 2 (Standard): Moderate-Risk Profile

- Loan length 37-48 months
- Debt-to-income ratio 30-40%
- Salary range: 40K-60K €
- Recommended Action: Approve with adjusted rates (5-8% APR) or require higher down payment

Tier 3 (Subprime): High-Risk Profile

- Loan length > 48 months
- Debt-to-income ratio > 40%
- No previous loan history
- Recommended Action: Manual review required; consider co-signer or collateral enhancement

11.5.2 Operational Efficiency Gains

The 94.1% accuracy enables automated decision-making for 85-90% of applications, reducing:

- Manual underwriting time by 60-70%
- Processing costs by approximately €15-20 per application
- Decision turnaround time from 3-5 days to <24 hours

11.5.3 Portfolio Risk Management

The model's high recall (94.98%) ensures that 95% of potential defaulters are flagged, enabling proactive interventions:

- Early warning systems for at-risk accounts
- Targeted financial counseling programs
- Refinancing options before delinquency occurs

11.5.4 Regulatory Compliance

The interpretable feature importance rankings support explainable AI requirements under EU regulations:

- Clear documentation of decision factors (loan length, DTI ratio)
- Absence of protected characteristics (age, gender) in top predictors
- Audit trail for adverse action explanations

11.6 Model Limitations and Future Enhancements

Despite strong performance, several limitations warrant consideration:

- 1. Driver Tenure Marginal Impact:** The 9th feature showed limited incremental value, suggesting diminishing returns from additional demographic variables without richer behavioral data.
- 2. Static Risk Assessment:** Current model lacks temporal dynamics—incorporating payment history trends or macroeconomic indicators could improve predictive power.
- 3. Threshold Optimization:** The default 0.5 probability cutoff may not align with business objectives. Cost-sensitive learning could optimize the precision-recall tradeoff based on actual default losses.

Recommended Next Steps:

- Integrate real-time payment behavior data (first 3-6 months of loan performance)
- Explore ensemble stacking with neural networks for non-linear interaction capture

- Develop segment-specific models for different car value brackets or regional markets

Section 11 Summary: The 9-feature optimized model achieved 94.1% accuracy and 97.2% ROC-AUC, validating its readiness for production deployment. Loan length and debt-to-income ratio emerged as dominant predictors, enabling risk-based pricing strategies and automated underwriting workflows. While driver tenure provided marginal incremental value, the overall feature set demonstrates robust generalization and strong business applicability.

12 Business Implications and Strategic Recommendations

12.1 Operational Value and Implementation Strategy

The 8-feature enhanced model delivers immediate operational benefits for car banks through automated risk assessment and cost reduction. With 94.2% accuracy and <100ms inference time, the model enables real-time loan application evaluation, replacing manual underwriting processes that typically require 2-5 business days. This automation can reduce processing costs by €25-40 per application while improving consistency across decisions.

12.1.1 Risk-Based Pricing Framework

The model's probability scores enable sophisticated pricing strategies that align risk with return. Table XVII presents a recommended tiered pricing structure based on predicted default probabilities.

Table 51: Risk-Based Pricing Recommendation

Risk Tier	Default Probability	Recommended APR	Expected Volume
Tier 1 (Low)	0-10%	4.5-6.5%	35% of applicants
Tier 2 (Medium)	10-20%	6.5-9.5%	40% of applicants
Tier 3 (High)	20-35%	9.5-13.5%	20% of applicants
Tier 4 (Very High)	>35%	Decline or require co-signer	5% of applicants

This framework can improve portfolio yield by 0.8-1.2 percentage points while safely increasing approval rates in the medium-risk segment by 10-15% through appropriate pricing adjustments.

12.2 Key Risk Drivers and Policy Recommendations

Feature importance analysis reveals actionable insights for lending policy optimization:

12.2.1 Loan Length Management (Importance: 100.0)

- Loans exceeding 60 months show significantly elevated default risk
- **Recommendation:** Implement tiered pricing with 1.5-2.5 percentage point premiums for extended terms; offer 0.5% rate reduction incentives for 36-month vs. 60-month loans

12.2.1 Debt-to-Income Ratio Controls (Importance: 83.13)

- DTI ratios >40% exhibit exponential default risk increase
- **Recommendation:** Establish hard DTI cap at 45% for standard approvals; require manual review and additional collateral for applications exceeding this threshold

12.2.3 Payment Affordability Assessment (Importance: 17.92)

- Payment-to-income ratio provides incremental predictive power beyond DTI
- **Recommendation:** Display affordability metrics prominently in customer-facing interfaces to encourage self-selection and reduce adverse selection

12.2.4 Down Payment Optimization (Importance: 14.67)

Larger down payments correlate with lower default risk through increased borrower commitment

- **Recommendation:** Develop incentive programs encouraging higher down payments (e.g., rate reductions of 0.25% per 10% down payment increase)

12.3 Implementation Roadmap

We recommend a phased deployment approach to minimize operational risk:

Phase 1: Pilot Testing (Months 1-3)

- Deploy model in shadow mode alongside existing underwriting process
- Compare model predictions against actual outcomes for 1,000+ applications
- Calibrate probability thresholds based on institutional risk appetite

Phase 2: Hybrid Decision Framework (Months 4-6)

- Auto-approve applications with predicted default probability <10% and DTI <35%
- Auto-decline applications with predicted default probability >35%
- Route borderline cases (10-35% probability) to manual underwriting with model scores as decision support

Phase 3: Full Automation (Months 7-12)

- Expand auto-approval coverage to 60-70% of applications
- Implement continuous monitoring dashboard tracking approval rates, default rates, and model drift
- Establish monthly retraining schedule using recent performance data

Data Quality Enhancement

- Reduce missing data through mandatory application fields for salary_range and driver_tenure
- Integrate third-party data sources (e.g., credit bureaus, employment verification APIs) for applicants willing to share information
- Explore alternative data (utility payments, rent history) to expand addressable market

13 Limitations and Future Research Directions

13.1 Data and Methodological Limitations

While our model demonstrates strong predictive performance, several limitations must be acknowledged:

13.1.1 Data Constraints

- **Missing Data:** 19% of observations required imputation (salary_range: 13.6%, driver_tenure: 5.4%), introducing uncertainty despite robust Random Forest imputation achieving 84.88% accuracy
- **Temporal Scope:** Cross-sectional dataset lacks longitudinal tracking of loan performance over time, limiting ability to capture dynamic risk patterns
- **Feature Gaps:** Absence of traditional credit scores, regional economic indicators (local unemployment rates), and vehicle-specific information (car age, type, resale value) that could enhance predictions
- **Economic Context:** Data collected during stable economic conditions may not generalize to recession or high-inflation periods

13.1.2 Methodological Considerations

- **Model Interpretability:** XGBoost's ensemble structure creates challenges for individual prediction explanations, potentially hindering regulatory compliance and customer communication (mitigated through SHAP values)
- **Threshold Selection:** Our 0.5 probability threshold is arbitrary; optimal threshold requires business-specific cost-benefit analysis balancing false positive costs (rejected viable applicants) vs. false negative costs (loan defaults)
- **External Validity:** Model trained on one car bank's data may not generalize to other institutions with different customer bases, geographic markets, or lending policies

13.1.3 Business Implementation Constraints

- **System Integration:** Deploying ML model requires integration with legacy loan origination systems, involving significant IT investment and change management
- **Regulatory Compliance:** Financial services regulators may require extensive validation and fairness testing before production deployment
- **Fairness Risks:** Features like postal_code or employment_type may correlate with protected characteristics (race, gender), creating potential for indirect discrimination that requires ongoing monitoring

13.2 Future Research Directions

Building on our findings, we propose the following research avenues:

13.2.1 Model Enhancement

- **Advanced Architectures:** Evaluate deep learning models (TabNet, FT-Transformer) and survival analysis techniques (Cox proportional hazards) to predict time-to-default rather than binary outcomes
- **Explainability Integration:** Implement SHAP values systematically for local explanations and develop customer-facing explanation templates (e.g., "Increasing down payment by €2,000 would improve approval probability by 25%")
- **Ensemble Stacking:** Combine XGBoost with complementary algorithms to capture diverse risk patterns

13.2.2 Data Enrichment

- **Alternative Data Sources:** Incorporate behavioral data (digital footprint, psychometric assessments) and telematics data (driving behavior from connected cars) for dynamic risk re-assessment
- **Longitudinal Studies:** Follow loan cohorts over full repayment periods (3-7 years) to validate long-term predictive accuracy and model stability across economic cycles

- **Macroeconomic Integration:** Add time-varying economic indicators (unemployment rate, interest rate trends) to capture cyclical risk patterns

13.2.3 Fairness and Ethics

- **Algorithmic Fairness Audits:** Assess model performance across protected demographic groups (when legally permissible data is available) and implement fairness constraints (demographic parity, equalized odds) during training
- **Regulatory Alignment:** Ensure model meets EU AI Act requirements for high-risk AI systems in financial services, including documentation, validation, and ongoing monitoring standards
- **Bias Mitigation:** Develop preprocessing and postprocessing techniques to reduce discriminatory patterns while maintaining predictive performance

13.2.4 Business Applications

- **Dynamic Risk Management:** Develop early warning systems predicting default risk for existing loans based on payment patterns and life events; use reinforcement learning to optimize intervention timing
- **Customer Lifetime Value:** Extend modeling to predict repeat business probability and cross-sell opportunities, enabling personalized product offerings
- **Portfolio Optimization:** Apply modern portfolio theory to balance risk-return across loan book and optimize capital allocation

13.3 Concluding Remarks

This study demonstrates that machine learning can effectively address credit risk assessment challenges for car banks operating without traditional credit histories. Our 8-feature enhanced XGBoost model achieves 94.2% accuracy and 97.3% AUC-ROC, with strategic feature engineering contributing meaningful predictive power while maintaining operational feasibility.

Key Contributions:

1. **Methodological:** Validates Random Forest imputation effectiveness for financial data with 19% missingness; demonstrates XGBoost superiority over traditional logistic regression in non-traditional lending contexts
2. **Practical:** Provides replicable framework for feature engineering in credit risk applications; delivers production-ready model with <100ms inference time
3. **Business:** Identifies actionable risk drivers (loan length, DTI ratio) enabling risk-based pricing strategies; quantifies potential for 15-25% default rate reduction and €25-40 per application cost savings

Implementation Recommendation: Car banks should proceed with phased deployment beginning with shadow mode testing, while simultaneously investing in data quality improvements and fairness auditing capabilities. The combination of technological capability and responsible implementation will determine competitive success in the data-driven lending era.

While limitations exist—particularly regarding temporal generalization, external validity, and fairness considerations—our findings provide a solid foundation for automated lending decisions in non-traditional financial institutions. As automotive financing continues evolving with digital transformation and alternative data proliferation, machine learning will play an increasingly central role in credit risk management.